

Key Biting Specifications

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Edition 1, 2025

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Acknowledgements

Much of the information in this book was compiled as part of a previous project between me, David Lewis, and Brendan Herman-Sullivan. Brendan and I spent over a year combing through our available resources to piece together each key biting specification listed in this book. When we stopped, we had completed 97 to our satisfaction. As of this writing, there are slightly over 50 key biting specifications still “in progress” that I hope to complete for future editions. Nevertheless, without Brendan’s efforts, this book would not be possible.

Thank you to Breck Camp, Roger Weitzenkamp, A.J. Hoffman, Billy B. Edwards, and Jerome Andrews. Brendan and I sourced a tremendous amount of information from their publications and/or notes.

Finally, a special thank you to those in this industry whose friendship and mentorship along the way has made these sorts of projects tolerable: Jeff Moss, Tim Coleman, Chris Huffstetler, Joshua Sands, Bill Lynk, John Hubel, Jim Hancock, Mark Dawson, Mark Blum, and Avi Schenk.

-Tyler J. Thomas 1/6/25

Introduction

A key biting specification can be used to determine quite a lot from a key and/or cylinder. Measurements with a precise caliper or micrometer can reveal the proper pinning, the keying system, applicable key blank(s), increment size, and many other characteristics that are valuable to locksmiths in the field or in the shop. You are only limited by your imagination with ways to utilize and “translate” the information from these key biting specifications for your benefit.

Each key biting specification contained in this book will contain the following information:

- Spacing
- Root Depths
- MACS (if applicable)
- Included Angle
- Root Cut

When available, other information such as effective plug diameter, increment size, progression type, and corresponding pins/wafers will also be included.

All key biting specifications are listed by manufacturer and labelled according to industry accepted nomenclature, such as Best A2 or A4, or increment size, such as Yale .025" and Yale .018". Additionally, special information related to the key biting specification can be found in the “Notes” box.

Information related to various key code equipment can be found in the “Key Origination” box. The “Blackhawk” box will refer you to the corresponding Blackhawk code card. The

“CodeMax” block will refer you to the corresponding HPC DSD number. The “HPC Blitz” box will refer you to the corresponding HPC code card. If manufactured, the “HPC Punch” box will refer you to the corresponding HPC 1200PCH card. Finally, the “ITL” box will refer you to the corresponding ITL 9700 number.

Finally, assembling a comprehensive list for key bitting specifications is a very tedious and laborious process. Information for most key bitting specifications is readily available, but some information has been lost to time, manufacturer mergers or closings, and/or product discontinuation. Every effort to compile these key bitting specifications started with information sourced directly from the manufacturer. In instances where this was not possible, information was drawn from aftermarket code equipment manufacturers, such as HPC’s DSD list or ITL’s Operations Manual, and code books. Every effort was made to meticulously scrutinize and “double-check” all key bitting specifications for accuracy. With that said, I am human. If you find a discrepancy, please contact me so that it can be fixed for future editions: tyler@sec-mls.com

High Security

Abus Vitess

5

MACS: 6
Increment: .0137"
Progression: Two Step
E.P.D. .550"
Included Angle: 110°
Root Cut: .0078"

Key Origination	
Blackhawk	
CodeMax	
HPC Blitz	
HPC Punch	
ITL	

Root Depths	
Z	.3445"
Y	.3307"
X	.3169"
0	.3031"
1	.2864"
2	.2756"
3	.2618"
4	.2480"
5	.2343"
6	.2205"
7	.2067"
8	.1929"
9	.1791"

Spacing	
1	.148"
2	.297"
3	.445"
4	.593"
5	.740"
6	.888"

Notes
1) Read and cut bow-to-tip. 2) Only depths 0 through 9 are allowed on cut 6. 3) Position 2 can only use depths 1 through 9. 4) Position 1 is fitted with the "Intop" pin and should not be altered nor master pinned. 5) Abus recommends not using master pins smaller than 4.

T.F.C: .1476"

B.C.C: .1476"

	Z	Y	X	0	1	2	3	4	5	6
Bottom Pins	.1980"	.2118"	.2256"	.2394"	.2531"	.2669"	.2807"	.2945"	.3083"	.3220"
Top Pins*	.197"									
	7	8	9							
Bottom Pins	.3358"	.3496"	.3634"							
Top Pins*	.197"									

	1	2	3	4	5	6	7	8	9	10
Master Pins	-	.0275"	.041"	.055"	.069"	.083"	.096"	.110"	.124"	.138"
	11	12	13							
	.125"	.165"	.179"							

*Steel .197" pins should be used in chambers 1 and 3, .177" brass spool pins should be used in chambers 2 and 4, and steel barrel pins should be used in chambers 5 and 6. Short brass or steel pins should be used in chambers 3, 5, and 6 when paired with size 8 or 9 bottom pins.

Notes:

ASSA Twin Series (New Style)

6

MACS: 5
Increment: .024"
Progression: Single Step
E.P.D. .509"
Included Angle: 90°
Root Cut: .032"

Key Origination	
Blackhawk	
CodeMax	DSD #96
HPC Blitz	CEX1
HPC Punch	
ITL	41

Root Depths	
1	.1587"
2	.1823"
3	.2059"
4	.2295"
5	.2531"
6	.2768"
7	.3004"
8	.3240"
9	.3476"

Spacing	
1	.2638"
2	.4134"
3	.5630"
4	.7126"
5	.8622"
6	1.0118"

Notes	
1) Read and cut tip-to-bow.	
2) Cuts range from 1 (deepest) to 9 (shallowest).	

T.F.C: .2638"

B.C.C: .1496"

	1	2	3	4	5	6	7	8	9	
Bottom Pins	.3503"	.3267"	.3031"	.2795"	.2559"	.2322"	.2086"	.1850"	.1614"	
Spool Top Pins	A = .157"		B = .196"		C = .236"		D = .276"			

Master and Build-Up Pins	1	2	3	4	5	6	7	8	9	10
	.0236"	.0472"	.0708"	.0944"	.1181"	.1417"	.1652"	.1889"	.2125"	.2361"
	11	12	13	14	15	16	17			
	.2597"	.2833"	.3069"	.3305"	.3541"	.3777"	.4013"			

		2	3	4	5	6	7	8	9	10
Hollow Top Pins		.0472"	.0708"	.0944"	.1181"	.1417"	.1652"	.1889"	.2125"	.2361"

Notes:

Emhart High Security - Z and DH Class (System 70)

7

MACS:	Varies
Increment:	.028"
Progression:	Single Step
E.P.D.	.552"
Included Angle:	90°
Root Cut:	.010"

Key Origination	
Blackhawk	
CodeMax	DSD 3023
HPC Blitz	CX1
HPC Punch	
ITL	162

Root Depths	
2	.305"
3	.277"
4	.249"
5	.221"
6	.193"

Spacing	
1	.250"
2	.406"
3	.562"
4	.718"
5	.874"
6	1.030"

5 ½ or %	.207"
----------	-------

T.F.C: .250"

B.C.C: .156"

Notes
1) Read and cut bow-to-tip.
2) Cut angles are Left (L or 1, -20°) or Right (R or 2, +20°).
3) All Emhart High Security cylinders are 6 pin and use .552" plugs.
4) MACS is 3 when adjacent cuts are different, 4 when the same.
5) Emhart High Security is compatible with standard Corbin Russwin Z and DH Class (System 70) but it does not use depth 1.
6) Depth 5 ½ or % is only used for construction keying.

High Security Pins

	2	3	4	5	6					
Bottom Pins	.242"	.270"	.298"	.326"	.354"					
Master Pins	.097"	.125"	.153"							
Top Pins	.193"			.158"						

*Pin lengths include the interlocking tabs.

**Master Pins have three angles, #1 is Left, #2 is Right, #3 is Same.

***High Security Top Pins are coded as 1 for .193" and 2 for .158".

Conventional Pins (for IC control chambers and 6th chamber of Brink and Blockout cylinders)

	1	2	3	4	5	6				
Bottom Pins		.231"	.260"	.288"	.316"	.344"				
Master Pins	.028"	.056"	.084"	.112"						
IC Top Pins		.198"	.171"	.142"	.114"	.087"				

* Top pin length is .171" for 6th chamber of Brink and Blockout cylinders.

	-4	-3	-2	-1	0	+1	+2	+3	+4	
Build-Up Pins	.030"	.058"	.087"	.114"	.142"	.171"	.198"	.226"	.253"	

Notes:

Corbin Russwin Access 3 - AP

8

MACS:	4	Key Origination		Root Depths		Spacing	
Increment:	.030"	Blackhawk		1	.271"	1	.244"
Progression:	Single Step	CodeMax	DSD #391	2	.241"	2	.414"
E.P.D.	.509"	HPC Blitz	C391	3	.211"	3	.584"
Included Angle:	90°	HPC Punch		4	.181"	4	.754"
Root Cut:	.054"	ITL		5	.151"	5	.924"
				6	.121"	6	1.094"

Notes	
1) Read and cut bow-to-tip.	
2) When using construction master keying, it is recommended that a depth of 6 be used for the CMK and a depth of 5 be used for the TMK in the lost wafer chamber.	

T.F.C: .244"

B.C.C: .170"

	1	2	3	4	5	6	7	8	9	
Bottom Pins	.231"	.261"	.291"	.321"	.351"	.381"				
Master Pins	.030"	.060"	.090"	.120"	.150"					
Top Pins	.030"	.060"	.090"	.120"	.150"	.180"	.210"	.240"	.270"	

Bump Resistant	1	2	3	4	5	6	7	8	9	
Top Pins				.120"	.150"	.180"	.210"	.240"	.270"	

	1	2	3	4	5	6	7	8	9	
Build-Up Pins (LFIC)	.030"	.060"	.090"	.120"	.150"	N/A	.210"	.240"	.270"	
	10	11								
	.300"	.330"								

*Access 3 cylinders have constant plug totals. For fixed cores and non-control chambers in LFIC stack height is 10, for LFIC control chambers stack height is 13.

Notes:

Corbin Russwin Access 3 - AS & AHS

9

MACS: 4
Increment: .030"
Progression: Single Step
E.P.D. .509"
Included Angle: 86°
Root Cut: .015"

Key Origination	
Blackhawk	
CodeMax	DSD #392
HPC Blitz	C392
HPC Punch	
ITL	

Root Depths*		
Number	Actual	Reference
1	.271"	.259"
2	.241"	.229"
3	.211"	.199"
4	.181"	.169"
5	.151"	.139"
6	.121"	.109"

Spacing	
1	.244"
2	.414"
3	.584"
4	.754"
5	.924"
6	1.094"

Notes
1) Read and cut bow-to-tip.
2) AS & AHS bottom bins are chisel tipped and rotate up to 20° from perpendicular.
3) Actual depths are as measured with a micrometer.

	Left	Center	Right
Cut Angles	-20°	0°	+20°

T.F.C: .244"

B.C.C: .170"

	1	2	3	4	5	6	7	8	9	
Bottom Pins	.231"	.261"	.291"	.321"	.351"	.381"				
Master Pins	.030"	.060"	.090"	.120"	.150"					
Top Pins	.030"	.060"	.090"	.120"	.150"	.180"	.210"	.240"	.270"	

	1	2	3	4	5	6	7	8	9	
Control Drivers	.030"	.060"	.090"	.120"	.150"	N/A	.210"	.240"	.270"	
(LFIC)	10	11								
	.300"	.330"								

*Access 3 cylinders have constant plug totals. For fixed cores and non-control chambers in LFIC stack height is 10, for LFIC control chambers stack height is 13.

Notes:

Corbin Russwin Pyramid

10

MACS: 7
Increment: .016"
Progression: Two Step
E.P.D. .466"
Included Angle: 90°
Root Cut: .053"

Key Origination	
Blackhawk	
CodeMax	DSD #17
HPC Blitz	CHS1
HPC Punch	PCH-1011
ITL	605

Root Depths	
0	.327"
1	.311"
2	.295"
3	.280"
4	.264"
5	.248"
6	.232"
7	.217"
8	.201"
9	.185"

Spacing	
1	.198"
2	.341"
3	.482"
4	.624"
5	.766"
6	.908"
7	1.049"

T.F.C: .198"

B.C.C: .1415"

Notes
1) Read and cut bow-to-tip.
2) Side Pin chambers are numbered bow-to-tip; right side 1-4, left side 6-9.
3) Pin diameters are .112".

	0	1	2	3	4	5	6	7	8	9
Bottom Pins	.139"	.155"	.171"	.187"	.202"	.218"	.234"	.250"	.265"	.281"
Master Pins			.031"	.047"	.063"	.078"	.094"	.110"	.126"	.141"
IC Top Pins	.189"	.173"	.157"	.141"	.126"	.110"	.094"	.078"	.063"	.047"

	-8	-7	-6	-5	-4	-3	-2	-1	0	+1
Build-Up Pins	.031"	.047"	.063"	.078"	.094"	.110"	.126"	.141"	.157"	.173"
	+2	+3	+4	+5	+6	+7	+8	+9		
	.189"	.204"	.220"	.236"	.252"	.267"	.283"	.299"		

*Conventional (Fixed Core): Use Top Pin A with plug totals 5,6,7,8,9. Use Top Pin C with plug totals 0,1,2,3,4.

*IC Non-Control Chambers: Use Top Pin B with plug totals 5,6,7,8,9. Use Top Pin C with plug totals 0,1,2,3,4.

	A	B	C
Top Pins	.141"	.204"	.236"

Notes:

Kaba Peaks 140 - Family A2

11

MACS: 8
Increment: .0125"
Progression: Two Step
E.P.D. Varies
Included Angle: 90°
Root Cut: .046"

Key Origination	
Blackhawk	
CodeMax	DSD #608
HPC Blitz	CPKS1
HPC Punch	
ITL	506

Root Depths	
0	.3187"
1	.3062"
2	.2937"
3	.2812"
4	.2687"
5	.2562"
6	.2437"
7	.2312"
8	.2187"
9	.2062"

Spacing	
1	.136"
2	.276"
3	.416"
4	.556"
5	.696"
6	.836"

T.F.C: .136"

B.C.C: .140"

Notes
1) Keys are cut and read tip-to-bow.
2) E.P.D. for SFIC is .425", everything else is .488".
3) In position next to Peak, cuts deeper than a #6 require the use of an asymmetrical cutter, such as the #1011.
4) When master keying, the position closest to the Peak should be progressed last.
5) A stack height of 26 is standard for all Peaks cylinders except conventional mortise and rim cylinders (31) and SFIC (23).

	0A	1A	2A	3A	4A	5A	6A	7AS*	8AS*	9AS*
A Series Bottom Pins	.107"	.1195"	.132"	.1445"	.157"	.1695"	.182"	.1945"	.207"	.2195"

	0J	1J	2JS*	3JS*	4JS*	5J	6J	7J	8J	9J
J Series Bottom Pins	.1695"	.182"	.1945"	.207"	.2195"	.2325"	.2445"	.257"	.2695"	.282"

* "S" indicates spool type bottom pins.

** J Series bottom pins are for large diameter plugs only. 0J thru 4JS are the same as 5A thru 9AS, respectively.

*** Bottom pins are .003" shorter than aftermarket replacement pins.

	2B	3B	4B	5B	6BS*	7B	8BS*	9B	10BS*	11B
Master, Build-Up, & Top Pins	.025"	.0375"	.050"	.0625"	.075"	.0875"	.100"	.1125"	.125"	.1375"
	12B	13B	14B	15B	16B	17B	18B	19B		
	.150"	.1625"	.175"	.1875"	.200"	.2125"	.225"	.2375"		

* "S" indicates spool type top pins.

Notes:

Kaba Peaks 140 - Family A4

12

MACS: 4
Increment: .021"
Progression: Single Step
E.P.D. Varies
Included Angle: 90°
Root Cut: .046"

Key Origination	
Blackhawk	
CodeMax	DSD #612
HPC Blitz	
HPC Punch	
ITL	278

Root Depths	
0	.318"
1	.297"
2	.276"
3	.255"
4	.234"
5	.213"

Spacing	
1	.136"
2	.276"
3	.416"
4	.556"
5	.696"
6	.836"

Notes
1) Keys are cut and read tip-to-bow.
2) E.P.D. for SFIC is .425", everything else is .488".
3) In position next to Peak, cuts deeper than a #3 require the use of an asymmetrical cutter, such as the #1011.
4) When master keying, the position closest to the Peak should be progressed last.
5) A stack height of 15 is standard for all Peaks cylinders except conventional mortise and rim cylinders (18) and SFIC (14).

T.F.C: .136"

B.C.C: .140"

	0E	1E	2E	3E	4ES*	5ES*				
E Series Bottom Pins	.107"	.128"	.149"	.1695"	.191"	.212"				

	0K	1KS*	2KS*	3K	4K	5K				
K Series Bottom Pins	.1695"	.191"	.212"	.2325"	.254"	.275"				

* "S" indicates spool type bottom pins.

** K Series bottom pins are for large diameter plugs only. 1KS and 2KS are the same as 4ES and 5ES, respectively.

*** Bottom pins are .003" shorter than aftermarket replacement pins.

	1F	2F	3F	4FS*	5FS*	6FS*	7F	8F	9F	10F
Master, Build-Up, & Top Pins	.021"	.042"	.063"	.084"	.105"	.125"	.147"	.168"	.189"	.210"
	11F									
	.231"									

* "S" indicates spool type top pins.

Notes:

Kaba Peaks 150 - Family A2

13

MACS: 9
Increment: .0125"
Progression: Two Step
E.P.D. .425"
Included Angle: 90°
Root Cut: .046"

Key Origination	
Blackhawk	
CodeMax	See notes
HPC Blitz	CPKS2
HPC Punch	
ITL	See notes

Root Depths	
0	.3187"
1	.3062"
2	.2937"
3	.2812"
4	.2687"
5	.2562"
6	.2437"
7	.2312"
8	.2187"
9	.2062"

Spacing	
1	.086"
2	.236"
3	.386"
4	.536"
5	.686"
6	.836"
7	.986"

T.F.C: .086"
B.C.C: .150"

Notes
1) Keys are cut and read tip-to-bow.
2) In position next to Peak, cuts deeper than a #6 require the use of an asymmetrical cutter, such as the #1011.
3) When master keying, the position closest to the Peak should be progressed last.
4) For Codemax, use DSD #606 for 6 pin, DSD #607 for 7 pin.
5) For ITL, use ITL# 505 for 6 pin, ITL# 279 for 7 pin.

	0A	1A	2A	3A	4A	5A	6A	7AS*	8AS*	9AS*
A Series Bottom Pins	.107"	.1195"	.132"	.1445"	.157"	.1695"	.182"	.1945"	.207"	.2195"

* "S" indicates spool type bottom pins.

** Bottom pins are .003" shorter than aftermarket replacement pins.

	2B	3B	4B	5B	6BS*	7B	8BS*	9B	10BS*	11B
Master, Build-Up, & Top Pins	.025"	.0375"	.050"	.0625"	.075"	.0875"	.100"	.1125"	.125"	.1375"
	12B	13B	14B	15B	16B	17B	18B	19B		
	.150"	.1625"	.175"	.1875"	.200"	.2125"	.225"	.2375"		

* "S" indicates spool type top pins.

Notes:

Kaba Peaks 150 - Family A4

14

MACS: 5
Increment: .021"
Progression: Single Step
E.P.D. .425"
Included Angle: 90°
Root Cut: .046"

Key Origination	
Blackhawk	
CodeMax	See notes
HPC Blitz	
HPC Punch	
ITL	See notes

Root Depths	
0	.318"
1	.297"
2	.276"
3	.255"
4	.234"
5	.213"

Spacing	
1	.086"
2	.236"
3	.386"
4	.536"
5	.686"
6	.836"
7	.986"

Notes
1) Keys are cut and read tip-to-bow.
2) In position next to Peak, cuts deeper than a #3 require the use of an asymmetrical cutter, such as the #1011.
3) When master keying, the position closest to the Peak should be progressed last.
4) For Codemax, use DSD #615 for 6 pin, DSD #616 for 7 pin.
5) For ITL, use ITL# 277 for 6 pin, ITL# 276 for 7 pin.

T.F.C: .086"

B.C.C: .150"

	0E	1E	2E	3E	4ES*	5ES*				
E Series Bottom Pins	.107"	.128"	.149"	.1695"	.191"	.212"				

	0K	1KS*	2KS*	3K	4K	5K				
K Series Bottom Pins	.1695"	.191"	.212"	.2325"	.254"	.275"				

* "S" indicates spool type bottom pins.

** Bottom pins are .003" shorter than aftermarket replacement pins.

Master, Build-Up, & Top Pins	1F	2F	3F	4FS*	5FS*	6FS*	7F	8F	9F	10F
	.021"	.042"	.063"	.084"	.105"	.125"	.147"	.168"	.189"	.210"
	11F									
	.231"									

* "S" indicates spool type top pins.

Notes:

Medeco Biaxial

15

MACS: Varies; see table.

Increment: .025"

Progression: Single Step

Included Angle: 86°

Biaxial Angles		
Fore	Original	Aft
K	L	M
B	C	D
Q	R	S

Biaxial MACS Violations							
Tip Side							
Bow Side		K	M	B	D	Q	S
	K	3	4	3	4	3	4
	M	2	3	2	3	2	3
	B	3	4	3	4	3	4
	D	2	3	2	3	2	3
	Q	3	4	3	4	3	4
	S	2	3	2	3	2	3

Bottom Pins*		
Pin	Reference Length	Measured
1	.239"	.227"
2	.264"	.252"
3	.289"	.277"
4	.314"	.302"
5	.339"	.327"
6	.364"	.352"

Spacing		
	Fore	Aft
1	.213"	.275"
2	.383"	.445"
3	.553"	.615"
4	.728"	.785"
5	.893"	.955"
6	1.063"	1.125"

Master Pins	
Pin	Length
1	.025"
2	.050"
3	.075"
4	.100"
5	.125"

Root Depths	
1	.264"
2	.239"
3	.214"
4	.189"
5	.164"
6	.139"

Top Pins**		
Pin	Length	Stack Height
1	.270"	.509"
2	.240"	.504"
3	.210"	.499"
4	.210"	.524"
5	.180"	.519"
6	.150"	.514"

* Reference length measures from the top of the pin to the theoretical chisel point. Actual measured length with a standard micrometer will be approximately .012" less than reference length.

** Biaxial cylinders do not have a constant stack height; stack height must be between .499" and .524". Top pin must match plug total.

Notes:

	Full Step	Half Step
MACS:	4	8
Increment:	.030"	.015"
Progression:	Single Step	Two Step
E.P.D.	.503"	.503"
Included Angle:	86°	86°
Root Cut:	.015"	.015"

Key Origination		
	Full Step	Half Step
Blackhawk	Card 191	
CodeMax	DSD #51	DSD #75
HPC Blitz	C36	CX4
HPC Punch		
ITL	298	302

Root Depths			
Full Step (.030")		Half Step (.015")	
1	.258"	0	.258"
2	.228"	1	.243"
3	.198"	2	.228"
4	.168"	3	.213"
5	.138"	4	.198"
6	.108"	5	.183"
		6	.168"
		7	.153"
		8	.138"
		9	.123"
		X	.108"

Notes
1) Read and cut bow-to-tip.
2) Cut angles are Left (L, -20°), Center (C, 0°), or Right(R, +20°).
3) Medeco pins are .135" diameter with a 85° angle tip.
4) Medeco pins will measure .012" shorter than listed and cut depths will measure .008" shallower when using standard calipers.
5) The first position should not use cuts deeper .138" to maintain key strength.

Spacing	
1	.244"
2	.414"
3	.584"
4	.754"
5	.924"
6	.1094"

T.F.C: .244"
B.C.C: .170"

	0	1	2	3	4	5	6	7	8	9	X
Full Step Pins	.080"	.110"	.140"	.170"	.200"	.230"					
Half Step Pins	.080"	.095"	.110"	.125"	.140"	.155"	.170"	.185"	.200"	.215"	.230"

	1&2	2&3	3&4	4&5	5&6						
Master Pins	.110"	.140"	.170"	.200"	.230"						

* 75 series master pins are only available in Full Step (.030") increments.

** 75 Series cam locks are mastered by adding a second hole for the sidebar at the necessary height and angle to match the position used for both cuts.

*** 75 series cam lock master pins are available in the following angle combinations: L & L, L & C, L & R, C & L, C & C, C & R, R & L, R & C, R & R.

Notes:

Medeco Original - Driverless, Full-Step

17

MACS: 4
Increment: .030"
Progression: Single Step
E.P.D. .503"
Included Angle: 86°
Root Cut: .015"

Key Origination	
Blackhawk	Card 191
CodeMax	DSD #51
HPC Blitz	C36
HPC Punch	
ITL	298

Root Depths	
1	.198"
2	.168"
3	.138"
4	.108"

Spacing		
	Old 60	Thick 60
1	.216"	.244"
2	.386"	.414"
3	.556"	.584"
4	.726"	.754"
5	.896"	.924"

Notes
1) Read and cut bow-to-tip.
2) Cut angles are Left (L, -20°), Center (C, 0°), or Right (R, +20°).
3) Old Style 60 series keys use .216" T.F.C., .170" B.C.C.
4) Medeco pins are .135" diameter with a 85° angle tip
5) Medeco pins can measure .012" shorter and cut depths .008" less than listed when using standard calipers.

T.F.C: .216" .244"

B.C.C: .170" .170"

	1	2	3	4	5	6				
Bottom Pins (60 series)	.090"	.120"	.150"	.180"						

	1 & 3	1 & 4	2 & 4							
Master Pins (60 series)	.140"	.170"	.170"							

* 60 Series cam locks are mastered by adding a second hole for the sidebar at the necessary height and angle to match the position used for both cuts.

60 Series master pins are available in the following depth combinations: 1 & 3, 1 & 4, 2 & 4.

** 60 series cam lock master pins are available in the following angle combinations: L & L, L & C, L & R, C & L, C & C, C & R, R & L, R & C, R & R.

Notes:

Medeco Original - Driverless, Half-Step

18

MACS: 8
Increment: .015"
Progression: Two-Step
E.P.D. .503"
Included Angle: 86°
Root Cut: .015"

Key Origination	
Blackhawk	
CodeMax	DSD-75
HPC Blitz	CX4
HPC Punch	
ITL	302

Root Depths	
4	.198"
5	.183"
6	.168"
7	.153"
8	.138"
9	.123"
X	.108"

Spacing		
	Old 60	Thick 60
1	.216"	.244"
2	.386"	.414"
3	.556"	.584"
4	.726"	.754"
5	.896"	.924"

T.F.C.: .216" .244"
B.C.C.: .170" .170"

Notes
1) Read and cut bow-to-tip.
2) Cut angles are Left (L, -20°), Center (C, 0°), or Right(R, +20°).
3) Old Style 60 series keys use .216" T.F.C, .170" B.C.C.
4) Medeco pins are .135" diameter with a 85° angle tip.
5) Medeco pins will measure .012" shorter than listed and cut depths will measure .008" shallower when using standard calipers.
6) Cam & Switch locks (60-series) only use depths 4 to X.
7) The first position should not use cuts deeper .138" to maintain key strength.

	0	1	2	3	4	5	6	7	8	9	X
Bottom Pins (60 series)					.090"	.105"	.120"	.135"	.150"	.165"	.180"

	4 & 7	4 & 8	4 & 9	4 & X	5 & 8	5 & 9	5 & X	6 & 9	6 & X	7 & X
Master Pins (60 series)	.135"	.150"	.165"	.180"	.150"	.165"	.180"	.165"	.180"	.180"

* 60 Series cam locks are mastered by adding a second hole for the sidebar at the necessary height and angle to match the position used for both cuts.

** 60 series master key systems can use up to 2 different skews across levels. Skew is determined by bottom pin.

*** 60-series pins made after July 1980 have a shallow groove in place of a false hole

**** 60 series cam lock master pins are available in the following angle combinations: L & L, L & C, L & R, C & L, C & C, C & R, R & L, R & C, R & R.

Notes:

Medeco Original - Full-Step

19

MACS: 4
Increment: .030"
Progression: Single Step
E.P.D. .503"
Included Angle: 86°
Root Cut: .015"

Key Origination	
Blackhawk	Card 191
CodeMax	DSD #51
HPC Blitz	C36
HPC Punch	
ITL	298

Root Depths	
1	.258"
2	.228"
3	.198"
4	.168"
5	.138"
6	.108"

Spacing	
1	.244"
2	.414"
3	.584"
4	.754"
5	.924"
6	1.094"

Notes
1) Read and cut bow-to-tip.
2) Cut angles are Left (L, -20°), Center (C, 0°), or Right (R, +20°).
3) Old Style 60 series keys use .216" T.F.C, .170" B.C.C.
4) Medeco pins are .135" diameter with a 85° angle tip
5) Medeco pins can measure up to .012" shorter than listed.
6) B and C top pins only used for locks manufactured prior to May, 1977.

T.F.C: .244"

B.C.C: .170"

	1	2	3	4	5	6				
Bottom Pins	.245"	.275"	.305"	.335"	.365"	.395"				

	1	2	3	4	5					
Master Pins	.030"	.060"	.090"	.120"	.150"					

*32 series Interchangeable Core cylinders have a difference of .090" between the operating and control shearlines. Change keys should use depth

4, 5, or 6 and Control keys should use depths 1, 2, or 3 in control chamber positions.

*10-50 series master key systems must share cut angles (L,C,R) at all levels. Skew is determined by bottom pin.

	B	C	1	2	3	4	5	6		
Top Pins	.330"	.300"	.270"	.240"	.210"	.180"	.150"	.120"		
Mushroom Driver Pins			.270"	.240"	.210"	.180"				

* .270" and .240" standard top pins are drill resistant steel, Mushroom pins are only available in brass

*10 series cylinders made prior to May 1977 (sheet metal cover) should have a total pin stack height of .575" to maintain correct spring pressure.

*10 series cylinders made after May 1977 (set screw cap) should have a total pin stack height of .515" to maintain correct spring pressure.

*20 Series cylinders (component/key-in-knob type) should have a total pin stack height of .515" to maintain correct spring pressure.

*32 Series cylinders (Interchangeable Core) should have a total pin stack height of .515" to maintain correct spring pressure.

Notes:

Medeco Original - Half-Step

20

MACS: 8
Increment: .015"
Progression: Two Step
E.P.D. .503"
Included Angle: 86°
Root Cut: .015"

Key Origination	
Blackhawk	
CodeMax	DSD-75
HPC Blitz	CX4
HPC Punch	
ITL	302

Root Depths	
0	.258"
1	.243"
2	.228"
3	.213"
4	.198"
5	.183"
6	.168"
7	.153"
8	.138"
9	.123"
X	.108"

Spacing	
1	.244"
2	.414"
3	.584"
4	.754"
5	.924"
6	.1094"

T.F.C.: .244"

B.C.C.: .170"

Notes
1) Cut angles are Left (L, -20°), Center (C, 0°), or Right(R, +20°).
2) Old Style 60 series keys use .216" T.F.C, .170" B.C.C.
3) Medeco pins are .135" diameter with a 85° angle tip.
4) Medeco pins will measure .012" shorter than listed and cut depths will measure .008" shallower when using standard calipers.
5) B and C top pins only used for locks manufactured prior to May, 1977.
6) Cam & Switch locks (60-series) only use depths 4 to X.
7) The first position should not use cuts deeper than .138" to maintain key strength.

	0	1	2	3	4	5	6	7	8	9	X
Bottom Pins	.245"	.260"	.275"	.290"	.305"	.320"	.335"	.350"	.365"	.380"	.395"

	2	3	4	5	6	7	8	9	X	
Master Pins	.030"	.045"	.060"	.075"	.090"	.105"	.120"	.135"	.150"	

*32 series Interchangeable Core cylinders have a difference of .090" between the operating and control shearlines.

*10-50 series master key systems must share cut angles (L,C,R) at all levels. Skew is determined by bottom pin.

	0	1 & 2	3 & 4	5 & 6	7 & 8	9 & X				
Top Pins	.270"	.240"	.210"	.180"	.150"	.120"				
Mushroom Driver Pins	.270"	.240"	.210"	.180"						

* .270" and .240" standard top pins are drill resistant steel, Mushroom pins are only available in brass

*10 series cylinders made prior to May 1977 (sheet metal cover) should have a total pin stack height of .575" to maintain correct spring pressure.

*10 series cylinders made after May 1977 (set screw cap) should have a total pin stack height of .515" to maintain correct spring pressure.

*20 Series cylinders (component/key-in-knob type) should have a total pin stack height of .515" to maintain correct spring pressure.

*32 Series cylinders (Interchangeable Core) should have a total pin stack height of .515" to maintain correct spring pressure.

Notes:

Medeco X4

21

MACS: 9
Increment: .0125"
Progression: Two Step
E.P.D. Varies
Included Angle: 90°
Root Cut: .054"

Key Origination	
Blackhawk	
CodeMax	DSD #1200
HPC Blitz	C3
HPC Punch	P3
ITL	311

Root Depths	
0	.3180"
1	.3055"
2	.2930"
3	.2805"
4	.2680"
5	.2555"
6	.2430"
7	.2305"
8	.2180"
9	.2055"

Spacing (Shoulder)	
1	.195"
2	.345"
3	.495"
4	.645"
5	.795"
6	.945"
7	1.095"

Spacing (Tip)	
1	.090"
2	.240"
3	.390"
4	.540"
5	.690"
6	.840"
7	.990"

T.F.C: Varies

B.C.C: .150"

Notes
1) Read and cut tip-to-bow.
2) Originally sold as Keymark X4.
3) Tip stop used for SFIC.
4) Shoulder stop used for all other cores/cylinders.
5) SFIC, Corbin Russwin/Sargent LFIC, and cam locks have a plug diameter of .434".
6) Conventional cylinders and Schlage/Yale LFIC have a plug diameter of .511".
7) Pin stack total for SFIC is 23.
8) Pin stack total for non-SFIC is 19.

.434" Diameter

	0	1	2	3	4	5	6	7*	8*	9*
Bottom Pins	.110"	.1225"	.135"	.1475"	.160"	.1725"	.185"	.1975"	.210"	.2225"

* Available in spool variation.

.511" Diameter

	0	1	2	3	4	5	6	7*	8*	9*
Bottom Pins	.1860"	.1985"	.2110"	.2235"	.2360"	.2485"	.2610"	.2735"	.2860"	.2985"

* Available in spool variation.

	2	3	4	5	6*	7	8*	9	10*	11
Master, Build-Up, & Top Pins	.025"	.0375"	.050"	.0625"	.075"	.0875"	.100"	.1125"	.125"	.1375"
	12	13	14	15	16	17	18	19		
	.150"	.1625"	.175"	.1875"	.200"	.2125"	.225"	.2375"		

* Available in spool variation.

Notes:

Medeco3 (M3)

22

MACS: Varies; see table.

Increment: .025"

Progression: Single Step

Included Angle: 86°

Medeco3 Angles		
Fore	Original	Aft
K	L	M
B	C	D
Q	R	S

Side Biting Locations		
Biting	Change	Master
1	.6085"	.6085"
2	.5585"	.5585"
3	.5085"	.5085"
4	.4585"	.4585"
5	.4085"	.4085"
6	.3585"	.3585"

Medeco3 MACS Violations							
Tip Side							
Bow Side	K	M	B	D	Q	S	
	K	3	4	3	4	3	4
	M	2	3	2	3	2	3
	B	3	4	3	4	3	4
	D	2	3	2	3	2	3
	Q	3	4	3	4	3	4
	S	2	3	2	3	2	3

Bottom Pins*		
Pin	Reference Length	Measured
1	.239"	.227"
2	.264"	.252"
3	.289"	.277"
4	.314"	.302"
5	.339"	.327"
6	.364"	.352"

Spacing		
	Fore	Aft
1	.213"	.275"
2	.383"	.445"
3	.553"	.615"
4	.728"	.785"
5	.893"	.955"
6	1.063"	1.125"

Master Pins	
Pin	Length
1	.025"
2	.050"
3	.075"
4	.100"
5	.125"

Top Pins**		
Pin	Length	Stack Height
1	.270"	.509"
2	.240"	.504"
3	.210"	.499"
4	.210"	.524"
5	.180"	.519"
6	.150"	.514"

Root Depths	
1	.272"
2	.247"
3	.222"
4	.197"
5	.172"
6	.147"

* Reference length measures from the top of the pin to the theoretical chisel point. Actual measured length with a standard micrometer will be approximately .012" less than reference length.

** M3 cylinders do not have a constant stack height; stack height must be between .499" and .524". Top pin must match plug total.

Notes:

Medeco 4 (M4)

23

MACS: Varies; see table.

Increment: .025"

Progression: Single Step

Included Angle: 86°

Medeco 4 Angles		
Fore	Original	Aft
K	L	M
B	C	D
Q	R	S

SideCut Spacing and Depths		
Bitting/Position	Depth	Position
1	.084"	.329"
2	.064"	.499"
3	.044"	.669"
4	N/A	.839"

Medeco 4 MACS Violations							
Tip Side							
Bow Side		K	M	B	D	Q	S
	K	3	4	3	4	3	4
	M	2	3	2	3	2	3
	B	3	4	3	4	3	4
	D	2	3	2	3	2	3
	Q	3	4	3	4	3	4
	S	2	3	2	3	2	3

Spacing		
	Fore	Aft
1	.213"	.275"
2	.383"	.445"
3	.553"	.615"
4	.728"	.785"
5	.893"	.955"
6	1.063"	1.125"

Root Depths	
1	.272"
2	.247"
3	.222"
4	.197"
5	.172"
6	.147"

Bottom Pins*		
Pin	Reference Length	Measured
1	.239"	.227"
2	.264"	.252"
3	.289"	.277"
4	.314"	.302"
5	.339"	.327"
6	.364"	.352"

Master Pins	
Pin	Length
1	.025"
2	.050"
3	.075"
4	.100"
5	.125"

Top Pins**		
Pin	Length	Stack Height
1	.270"	.509"
2	.240"	.504"
3	.210"	.499"
4	.210"	.524"
5	.180"	.519"
6	.150"	.514"

* Reference length measures from the top of the pin to the theoretical chisel point. Actual measured length with a standard micrometer will be approximately .012" less than reference length.

** Medeco 4 cylinders do not have a constant stack height; stack height must be between .499" and .524". Top pin must match plug total.

*** 6 pin plugs utilize 4 sidepin chambers, 5 pin plugs utilize 3 sidepin chambers. The chamber which interacts with the shuttle pin will use a 0 pin. Pins 1 - 3 pair with their coded depth. Pin 4 has open positions for 1 & 2 coded depths, Pin 5 has open positions for coded depths 2 & 3.

Notes:

Mul-T-Lock - Classic and Interactive

24

MACS: See Notes

Increment: .0196"

Progression: Single Step

E.P.D. .3011"

Included Angle: 90°

Root Cut: .035"

Key Origination	
Blackhawk	
CodeMax	
HPC Blitz	
HPC Punch	
ITL	

Root Depths	
0/Z	n/a*
1/A	.0944"
2/B	.0748"
3/C	.0552"
4/D	.0354"
5	.0157"

Spacing	
1	.9642"
2	.7753"
3	.5864"
4	.3975"
5	.2086"

T.F.C: .2086"

B.C.C: .1889"

Notes										
1) Read bow to tip, cut tip to bow. 2) Depths Z and 0 are only used in Interactive. 3) The "Interactive Chamber" is located in the frontmost or second frontmost chamber of the cylinder and is populated to match the "Interactive Element". 4) The "Interactive Element" on a keyblank is in position 4 or 5. The element is a mechanical rocker capable of extending beyond the uncut depth of the blank. 5) Internal Bottom Pins may be no more than one depth longer or shorter than the accompanying External pin. There is no limit to depth changes between chambers. 6) The cylinder format determines which type of top pin is used; top pins are uniform in size within that cylinder format.										

Bottom Pins

	0*	1	2	3	4	5				
Internal	.187"	.2066"	.2263"	.246"	.2657"	.2854"				
	Z*	A	B	C	D					
External	.183"	.2047"	.2224"	.2421"	.2618"					

*0 and Z sized pins are only used for the Interactive Chamber as they are shorter than the plug chamber.

Master Pins

	1	2	3	4						
Internal	.0196"	.0393"	.059"	.0787"						
External	.0196"	.0393"	.059"							

	G/Gold	B/Black	S/Silver							
Interactive Element	A-0	Z-1	Z-0							

Notes:

Sargent Degree - DG1

25

MACS: 4
Increment: .030"
Progression: Single Step
E.P.D. .502"
Included Angle: 90°
Root Cut: .054"

Key Origination	
Blackhawk	
CodeMax	DSD #389
HPC Blitz	C389
HPC Punch	
ITL	

Root Depths	
1	.271"
2	.241"
3	.211"
4	.181"
5	.151"
6	.121"

Spacing	
1	.244"
2	.414"
3	.584"
4	.754"
5	.924"
6	1.094"
7*	1.264"

Notes
1) Read and cut bow-to-tip.
2) The 50-DG1 hotel mortise cylinder uses 6 pin change/master keys but with a #5 depth cut in what would be the 7th position at the tip of the key. 7th chamber always loaded with a #5 bottom pin and #8 driver.

T.F.C: .244"

B.C.C: .170"

	1	2	3	4	5	6	7	8	9	
Bottom Pins	.231"	.261"	.291"	.321"	.351"	.381"				
Master Pins	.030"	.060"	.090"	.120"	.150"					
Top Pins	.030"	.060"	.090"	.120"	.150"	.180"	.210"	.240"	.270"	

Bump Resistant	1	2	3	4	5	6	7	8	9	
Top Pins				.120"	.150"	.180"	.210"	.240"	.270"	

	1	2	3	4	5	6	7	8	9	
Control Drivers (LFIC)	.030"	.060"	.090"	.120"	.150"	N/A	.210"	.240"	.270"	
	10	11								
	.300"	.330"								

*Degree cylinders have constant plug totals. For fixed cores and non-control chambers in LFIC stack height is 10, for LFIC control chambers stack height is 13.

Notes:

Sargent Degree - DG2 & DG3

26

MACS: 4
Increment: .030"
Progression: Single Step
E.P.D. .502"
Included Angle: 86°
Root Cut: .015"

Key Origination	
Blackhawk	
CodeMax	DSD #390
HPC Blitz	C390
HPC Punch	
ITL	298

Root Depths*		
Number	Actual	Reference
1	.271"	.259"
2	.241"	.229"
3	.211"	.199"
4	.181"	.169"
5	.151"	.139"
6	.121"	.109"

Spacing	
1	.244"
2	.414"
3	.584"
4	.754"
5	.924"
6	1.094"

Notes
1) Read and cut bow-to-tip.
2) Chambers 1-5 use L, C, R angled pins. Chamber 6 uses same bottom pins (conical) as for DG1 cylinders.
3) Actual depths are as measured with a micrometer.
4) ITL uses 3M3 insert.

T.F.C: .244"

B.C.C: .170"

	Left	Center	Right
Cut Angles	-20°	0°	+20°

	1	2	3	4	5	6	7	8	9	
Bottom Pins	.231"	.261"	.291"	.321"	.351"	.381"				
Master Pins	.030"	.060"	.090"	.120"	.150"					
Driver Pins	.030"	.060"	.090"	.120"	.150"	.180"	.210"	.240"	.270"	

	1	2	3	4	5	6	7	8	9	
Control Drivers (LFIC)	.030"	.060"	.090"	.120"	.150"	N/A	.210"	.240"	.270"	
	10	11								
	.300"	.330"								

*Degree cylinders have constant plug totals. For fixed cores and non-control chambers in LFIC stack height is 10, for LFIC control chambers stack height is 13.

Notes:

Schlage Primus, Primus XP, Everest Primus

27

MACS: 7
Increment: 0.15"
Progression: Two Step
E.P.D. .500"
Included Angle: 100°
Root Cut: .031"

Key Origination	
Blackhawk	Card 198
CodeMax	DSD #60
HPC Blitz	C45
HPC Punch	P45
ITL	391

Root Depths	
0	.335"
1	.320"
2	.305"
3	.290"
4	.275"
5	.260"
6	.245"
7	.230"
8	.215"
9	.200"

Spacing	
1	.231"
2	.387"
3	.543"
4	.699"
5	.855"
6	1.011"

T.F.C: .231"

B.C.C: .156"

Notes
1) Read and cut tip-to-bow.
2) Applies to Primus, Primus XP, Everest Primus, Everest Primus XP, Everest 29 Primus, and Everest 29 Primus XP.
3) Finger Pin #7 is only used in XP systems.

Side Biting Spacing			
	Left	Center	Right
1	.277"	.309"	.341"
2	.433"	.465"	.497"
3	.589"	.621"	.653"
4	.745"	.777"	.809"
5	.901"	.933"	.965"

Side Biting Root Depths			
	Left	Center	Right
Extra Shallow	.072"	-	.072"
Shallow	.048"	.060"	.048"
Deep	.024"	.036"	.024"

Finger Pin Number			
	Left	Center	Right
Extra Shallow	-	-	7
Shallow	1	3	5
Deep	2	4	6

	0	1	2	3	4	5	6	7	8	9
Bottom Pins	.165"	.180"	.195"	.210"	.225"	.240"	.255"	.270"	.285"	.300"
Master Pins			.030"	.045"	.060"	.075"	.090"	.105"	.120"	
Top Pins	.235"				.200"			.165"		

Notes:

Schlage 29 SL Everest, Primus, Everest Primus

28

MACS: 8
Increment: 0.0125"
Progression: Two Step
E.P.D. N/A
Included Angle: 90°
Root Cut: .052"

Key Origination	
Blackhawk	
CodeMax	DSD #320
HPC Blitz	C116
HPC Punch	P45
ITL	435

Root Depths	
0	.3187"
1	.3062"
2	.2937"
3	.2812"
4	.2687"
5	.2562"
6	.2437"
7	.2312"
8	.2187"
9	.2062"

Spacing	
1	1.096"
2	.946"
3	.796"
4	.646"
5	.496"
6	.346"
7	.196"

T.F.C: .196"

B.C.C: .150"

Notes
1) Read and cut tip-to-bow.
2) Applies to Primus 29 SL, Primus XP 29 SL, Everest Primus 29 SL, Everest Primus XP 29 SL, Everest 29 SL, and Everest XP 29 SL.
3) Finger Pin #7 is only used in XP systems.

Side Biting Spacing			
	Left	Center	Right
1	.277"	.309"	.341"
2	.433"	.465"	.497"
3	.589"	.621"	.653"
4	.745"	.777"	.809"
5	.901"	.933"	.965"

Side Biting Root Depths			
	Left	Center	Right
Extra Shallow	.072"	-	.072"
Shallow	.048"	.060"	.048"
Deep	.024"	.036"	.024"

Finger Pin Number			
	Left	Center	Right
Extra Shallow	-	-	7
Shallow	1	3	5
Deep	2	4	6

	0	1	2	3	4	5	6	7	8	9
Master L-pins*	0,2	1,3	2,4	3,5	4,6	5,7	6,8	7,9	-	-
	0,4	1,5	2,6	3,7	4,8	5,9	-	-	-	-
	0,6	1,7	2,8	3,9	-	-	-	-	-	-
	0,8	1,9	-	-	-	-	-	-	-	-

*Non-master L-pins (1 notch) are available in steps 0-9, Mastered L-Pins (2 notches) are shown in the chart above.

Notes:

MACS: 9
Increment: .0125"
Progression: Two Step
E.P.D. Varies
Included Angle: 90°
Root Cut: .054"

Key Origination	
Blackhawk	
CodeMax	DSD# 99
HPC Blitz	CKYM1
HPC Punch	
ITL	311

Root Depths	
0	.1385"
1	.1260"
2	.1135"
3	.1010"
4	.0885"
5	.0760"
6	.0635"
7	.0510"
8	.0385"
9	.0260"

Spacing (Shoulder)	
1	.195"
2	.345"
3	.495"
4	.645"
5	.795"
6	.945"
7	1.095"

Notes
1) Read and cut tip-to-bow.
2) Tip stop used for SFIC. Shoulder stop used for all other cores.
3) Depths are measured from the ledge of the key, not the angled leg.
4) SFIC, Corbin Russwin/Sargent LFIC, and cam locks have a plug diameter of .434".
5) Conventional cylinders and Schlage/Yale LFIC have a plug diameter of .511".
7) Pin stack total for SFIC is 23.
8) Pin stack total for non-SFIC is 19.

T.F.C: Varies

B.C.C: .150"

Spacing (Tip)	
1	.090"
2	.240"
3	.390"
4	.540"
5	.690"
6	.840"
7	.990"

	0	1	2	3	4	5	6	7*	8*	9*
Bottom Pins	.110"	.122"	.135"	.147"	.160"	.172"	.185"	.197"	.210"	.222"

* Available in spool variation.

	2	3	4	5	6*	7	8*	9	10*	11
Master, Build-Up, & Top Pins	.025"	.037"	.050"	.062"	.075"	.087"	.100"	.112"	.125"	.137"
	12	13	14	15	16	17	18	19		
	.150"	.162"	.175"	.187"	.200"	.212"	.225"	.237"		

* Available in spool variation.

Notes:

Lever

Key Origination	
Blackhawk	
CodeMax	DSD #1574
HPC Blitz	
HPC Punch	
ITL	386

Spacing	
1	.167"
2	.222"
3	.271"
4	.319"
5	.367"
6	.415"
7	.470"
Throat	1.0875"

Cut Width	
1	.066"
2	.052"
3	.052"
4	.052"
5	.052"
6	.052"
7	.066"
Throat	.105"

Notes
1) Read and cut tip-to-bow.
2) Spacing is from front of tip-stop to center-of-cut.
3) Use DSD #1575 for throat cut, use modified horseshoe tip stop HPC HT-SD or equivalent.
4) For ITL, cuts 8 and 9 are throat cut.

Root Depths	
0	.314"
1	.298"
2	.278"
3	.258"
4	.238"
5	.218"
6	.198"
7	.178"
Throat	.235"

	Blanks
S&G	96
Ilco	1063E
Jet	S&G-3-NS

Blade Height: .315" Tip Height: .070"

Notes:

Key Origination	
Blackhawk	
CodeMax	DSD #556
HPC Blitz	CSD41
HPC Punch	
ITL	

Spacing	
1	.167"
2	.222"
3	.271"
4	.319"
5	.378"
Throat	1.0775"

Cut Width	
1	.066"
2	.052"
3	.052"
4	.052"
5	.066"
Throat	.105"

Notes
1) Read and cut tip-to-bow.
2) Spacing is from front of tip-stop to center-of-cut.
3) Use DSD #1556 for throat cut, use modified horseshoe tip stop HPC HT-SD or equivalent.
5) S&G factory standard locks will use only even numbered cuts on guard keys.

Root Depths	
0	.275"
1	.255"
2	.235"
3	.215"
4	.195"
5	.175"
6	.155"
7	.135"
Throat	.200"

	Blanks			
	4440	4442	4443	4444
S&G	87H	87H	9609	87H
Ilco	SG4/1387	SG4/1387	1362	SG4/1387

Blade Height*: .275" .275" .350"/.280" .275"

Tip Height: .080" .080" .070" .080"

*Where two heights are shown, the first number is the height closest to the bow and the second is the height closer to the tip.

Notes:

Key Origination	
Blackhawk	
CodeMax	DSD #555
HPC Blitz	CSD40
HPC Punch	
ITL	385

Spacing	
1	.167"
2	.222"
3	.271"
4	.319"
5	.367"
6	.422"
Throat	1.0875"

Cut Width	
1	.066"
2	.052"
3	.052"
4	.052"
5	.052"
6	.066"
Throat	.105"

Notes
1) Read and cut tip-to-bow.
2) Spacing is from front of tip-stop to center-of-cut.
3) Use DSD #1555 for throat cut., use modified horseshoe tip stop HPC HT-SD or equivalent.
4) For ITL, cuts 7 and 8 are throat cut.

Root Depths	
0	.314"
1	.298"
2	.278"
3	.258"
4	.238"
5	.218"
6	.198"
7	.178"
Throat	.235"

	Blanks			
	4440	4442	4443	4444
S&G	96	60	60	96
Ilco	1063E	1063B	1063B	1063E
Jet	S&G-3-NS	S&G-2-NS	S&G-2-NS	S&G-3-NS

Blade Height*: .315" .390"/.315" .390"/.315" .315"

Tip Height: .070" .070" .070" .070"

*Where two heights are shown, the first number is the height closest to the bow and the second is the height closer to the tip.

Notes:

Key Origination	
Blackhawk	
CodeMax	DSD #1583
HPC Blitz	
HPC Punch	
ITL	

Spacing	
1	.167"
2	.222"
3	.271"
4	.319"
5	.367"
6	.415"
7	.470"
Throat	1.0775"

Cut Width	
1	.066"
2	.052"
3	.052"
4	.052"
5	.052"
6	.052"
7	.066"
Throat	.105"

Notes
1) Read and cut tip-to-bow.
2) Spacing is from front of tip-stop to center-of-cut.
3) Use DSD #1584 for throat cut, use modified horseshoe tip stop HPC HT-SD or equivalent.
4) The protruding tip on guard keys should be cut to .087" tall.

Root Depths	
0	.275"
1	.255"
2	.235"
3	.215"
4	.195"
5	.175"
6	.155"
7	.135"
Throat	.235"

Blanks	
S&G	87H
Ilco	SG4/1387

Blade Height*: .275" Thickness: .080"

Notes:

Key Origination	
Blackhawk	
CodeMax	DSD #1574
HPC Blitz	
HPC Punch	
ITL	386

Spacing	
1	.167"
2	.222"
3	.271"
4	.319"
5	.367"
6	.415"
7	.470"
Throat	1.0875"

Cut Width	
1	.066"
2	.052"
3	.052"
4	.052"
5	.052"
6	.052"
7	.066"
Throat	.105"

Notes
1) Read and cut tip-to-bow.
2) Spacing is from front of tip-stop to center-of-cut.
3) Use DSD #1575 for throat cut., use modified horseshoe tip stop HPC HT-SD or equivalent.
4) For ITL, cuts 8 and 9 are throat cut.

Root Depths	
0	.314"
1	.298"
2	.278"
3	.258"
4	.238"
5	.218"
6	.198"
7	.178"
Throat	.235"

	Blanks
S&G	96
Ilco	1063E
Jet	S&G-3-NS

Blade Height: .315"

Thickness: .070"

Notes:

S&G 4231 Guard

36

Key Origination	
Blackhawk	
CodeMax	DSD #556
HPC Blitz	CSD41
HPC Punch	
ITL	

Spacing	
1	.167"
2	.222"
3	.271"
4	.319"
5	.378"
6	1.0775"

Cut Width	
1	.066"
2	.052"
3	.052"
4	.052"
5	.066"
Throat	.105"

Notes
1) Read and cut tip-to-bow.
2) Spacing is from front of tip-stop to center-of-cut.
3) Use DSD #1556 for throat cut, use modified horseshoe tip stop HPC HT-SD or equivalent.
4) The protruding tip on guard keys should be cut to .087" tall.

Root Depths	
0	.275"
1	.255"
2	.235"
3	.215"
4	.195"
5	.175"
6	.155"
7	.135"
Throat	.200"

	Blanks
S&G	SY3B
Ilco	887

Blade Height: .243" Tip Height: .063"

Notes:

Key Origination	
Blackhawk	
CodeMax	DSD #555
HPC Blitz	CSD40
HPC Punch	
ITL	385

Spacing	
1	.167"
2	.222"
3	.271"
4	.319"
5	.367"
6	.422"
Throat	1.0775"

Cut Width	
1	.066"
2	.052"
3	.052"
4	.052"
5	.052"
6	.066"
6	.105"

Notes
1) Read and cut tip-to-bow.
2) Spacing is from front of tip-stop to center-of-cut.
3) Use DSD #1555 for throat cut., use modified horseshoe tip stop HPC HT-SD or equivalent.
4) For ITL, cuts 7 and 8 are throat cut.

Root Depths	
0	.314"
1	.298"
2	.278"
3	.258"
4	.238"
5	.218"
6	.198"
7	.178"
Throat	.235"

	Blanks
S&G	96
Ilco	1063E
Jet	S&G-3-NS

Blade Height: .315" Tip Height: .070"

Notes:

Pin Tumbler

American-Junkunc Small Pin

39

MACS: 4
Increment: .0156"
Progression: Two Step
E.P.D. .394"
Included Angle: 90°
Root Cut: .052"

Key Origination	
Blackhawk	Card 24
CodeMax	DSD #45
HPC Blitz	C30
HPC Punch	P30
ITL	26

Root Depths	
1	.284"
2	.2684"
3	.2528"
4	.2372"
5	.2215"
6	.2059"
7	.1903"
8	.1747"

Spacing	
1	.156"
2	.281"
3	.406"
4	.531"
5	.656"
6	.781"

T.F.C: .156"

B.C.C: .125"

Notes
1) Master Lock uses the same pin sizes renumbered 0-7 instead of 1-8.
2) Bottom pins 3-8 are serrated, top pins are available serrated or serrated spool.
3) Pins are .094" diameter.

	1	2	3	4	5	6	7	8
Bottom Pins	.110"	.126"	.140"	.156"	.172"	.190'	.202"	.220"
Master Pins		.031"	.046"	.062"	.078'	.094"	.110"	
Top Pins	.140"							

Notes:

Arrow New Large Pin

40

MACS: 7
Increment: .014"
Progression: Single Step
E.P.D. .492"
Included Angle: 100°
Root Cut: .046"

Key Origination	
Blackhawk	Card 411
CodeMax	DSD #2
HPC Blitz	C2
HPC Punch	
ITL	27

Root Depths	
0	.312"
1	.298"
2	.284"
3	.270"
4	.256"
5	.242"
6	.228"
7	.214"
8	.200"
9	.186"

Spacing	
1	.265"
2	.420"
3	.575"
4	.730"
5	.885"
6	1.040"

T.F.C: .265"

B.C.C: .155"

Notes
1) Read and cut bow-to-tip.

	0	1	2	3	4	5	6	7	8	9
Bottom Pins	.180"	.194"	.208"	.222"	.236"	.250"	.264"	.278"	.292"	.306"
Master Pins			.028"	.042"	.056"	.070"	.084"	.098"	.112"	.126"
Top Pins	.155"	.155"	.155"	.155"	.155"	.155"	.155"	.155"	.155"	.155"

Notes:

BEST A2

41

MACS: 9
Increment: .0125"
Progression: Two Step
E.P.D. .428"
Included Angle: 90°
Root Cut: .054"

Key Origination	
Blackhawk	
CodeMax	DSD #3
HPC Blitz	C3
HPC Punch	P3
ITL	54

Root Depths	
0	.318"
1	.3055"
2	.293"
3	.2805"
4	.268"
5	.2555"
6	.243"
7	.2305"
8	.218"
9	.2055"

Spacing	
1	.088"
2	.238"
3	.388"
4	.538"
5	.688"
6	.838"
7	.988"

Notes
1) Read and cut tip-to-bow.
2) 19A pins used for blockout cores.
3) 1B used in RARE master key systems incorporating existing keys.

T.F.C: .088"
B.C.C: .150"

	0A	1A	2A	3A	4A	5A	6A	7A	8A	9A
Bottom Pins	.110"	.1225"	.135"	.1475"	.160"	.1725"	.185"	.1975"	.210"	.2225"
	19A									
	.3475"									

	2B	3B	4B	5B	6B	7B	8B	9B	10B	11B
Master, Build-Up, & Top Pins	.025"	.0375"	.050"	.0625"	.075"	.0875"	.100"	.1125"	.125"	.1375"
	12B	13B	14B	15B	16B	17B	18B	19B		
	.150"	.1625"	.175"	.1875"	.200"	.2125"	.225"	.2375"		
	1B									
	.0125"									

Notes:

BEST A3

42

MACS: 6
Increment: .018"
Progression: Single Step
E.P.D. .428"
Included Angle: 90°
Root Cut: .054"

Key Origination	
Blackhawk	
CodeMax	DSD #302
HPC Blitz	CX10
HPC Punch	
ITL	55

Root Depths	
0	.318"
1	.300"
2	.282"
3	.264"
4	.246"
5	.228"
6	.210"

Spacing	
1	.088"
2	.238"
3	.388"
4	.538"
5	.688"
6	.838"
7	.988"

Notes
1) Read and cut tip-to-bow.
2) Discontinued; only sold to service existing systems.
3) 13C pins used for blackout cores.
4) MACS is technically 6 as there are no violations in system.

T.F.C: .088"

B.C.C: .150"

	0C	1C	2C	3C	4C	5C	6C			
Bottom Pins	.110"	.128"	.146"	.164"	.182"	.200"	.218"			
	13C									
	.343"									

	1D	2D	3D	4D	5D	6D	7D	8D	9D	10D
Master, Build-Up, & Top Pins	.018"	.036"	.054"	.072"	.090"	.108"	.126"	.144"	.162"	.180"
	11D	12D	13D							
	.198"	.216"	.234"							

Notes:

BEST A4

43

MACS: 5
Increment: .021"
Progression: Single Step
E.P.D. .428"
Included Angle: 90°
Root Cut: .054"

Key Origination	
Blackhawk	
CodeMax	DSD #303
HPC Blitz	CX11
HPC Punch	
ITL	56

Root Depths	
0	.318"
1	.297"
2	.276"
3	.255"
4	.234"
5	.213"

Spacing	
1	.088"
2	.238"
3	.388"
4	.538"
5	.688"
6	.838"
7	.988"

Notes
1) Read and cut tip-to-bow.
2) 11E pins used for blockout cores.
3) MACS is technically 5 as there are no violations in system.

T.F.C: .088"

B.C.C: .150"

Bottom Pins	0E	1E	2E	3E	4E	5E				
	.110"	.131"	.152"	.173"	.194"	.215"				
	11E									
	.340"									

Master, Build-Up, & Top Pins	1F	2F	3F	4F	5F	6F	7F	8F	9F	10F
	.021"	.042"	.063"	.084"	.105"	.126"	.147"	.168"	.189"	.210"
	11F									
	.231"									

Notes:

Corbin Russwin Z and DH Class (System 70)

44

MACS: 4
Increment: .028"
Progression: Single Step
E.P.D. Varies
Included Angle: 90°
Root Cut: .053"

Key Origination	
Blackhawk	Card 439
CodeMax	Varies
HPC Blitz	CX6A
HPC Punch	
ITL	122

Root Depths	
1	.339"
2	.311"
3	.283"
4	.255"
5	.227"
6	.199"

Spacing	
1	.250"
2	.406"
3	.562"
4	.718"
5	.874"
6	1.030"
7	1.186"

5 ½ or %	.213"
----------	-------

T.F.C: .250"

B.C.C: .156"

Notes
1) Read and cut bow-to-tip.
2) .509" diameter plugs have EPD of .504", .552" are .546"
3) For CodeMax, use DSD #614 for IC, DSD #3614 for Non-IC.
4) Use .320" top pin for PL5000 padlock / to block master ring.
5) 5 ½ or % cut used on construction master keys.

.509" Diameter

	1	2	3	4	5	6				
Bottom Pins	.160"	.189"	.217"	.245"	.273"	.301"				
Master Pins	.028"	.056"	.084"	.112"	.140"					
IC Top Pins	.192"	.163"	.135"	.107"	.080"	.080"				

* Standard top pin length is .171".

** Use .247" top pin for non-control chambers.

	-4	-3	-2	-1	0	+1	+2	+3	+4	+5
Build-Up Pins	.051"	.080"	.107"	.135"	.163"	.192"	.218"	.247"	.275"	.303"

.552" Diameter

	1	2	3	4	5	6				
Bottom Pins	.203"	.231"	.260"	.288"	.316"	.344"				
Master Pins	.028"	.056"	.084"	.112"	.140"					

* Standard top pin length is .171".

	-4	-3	-2	-1	0	+1	+2	+3	+4	+5
Build-Up Pins	.030"	.058"	.087"	.114"	.142"	.171"	.198"	.226"	.253"	.282"

Notes:

Corbin X Class (Pre-System 70)

45

MACS:	8	Key Origination		Root Depths		Spacing	
Increment:	.014"	Blackhawk	Card 437	1	.333"	1	.197"
Progression:	Two Step	CodeMax	Varies	2	.319"	2	.353"
E.P.D.	Varies	HPC Blitz	C13	3	.305"	3	.509"
Included Angle:	90°	HPC Punch		4	.291"	4	.665"
Root Cut:	.053"	ITL	111	5	.277"	5	.821"
Notes 1) Read and cut tip-to-bow. 2) .509" diameter plugs have EPD of .504", .552" are .546" 3) For CodeMax, use DSD #16 for IC, DSD #3016 for Non-IC. 4) Use .320" top pin for PL5000 padlock / to block master ring. 5) 11 cut used on some construction master keys.				6	.263"	6	.977"
				7	.249"	7	1.133"
				8	.235"	T.F.C: .197"	
				9	.221"	B.C.C: .156"	
				0	.207"		
				11*	.193"		

.509" Diameter

	1	2	3	4	5	6	7	8	9	0
Bottom Pins	.171"	.186"	.198"	.213"	.228"	.241"	.256"	.269"	.283"	.297"
Master Pins		.028"	.042"	.056"	.070"	.084"	.098"	.112"	.126"	
IC Top Pins	.192"	.177"	.163"	.149"	.135"	.12"	.107"	.093"	.080"	.066"

* Standard top pin length is .171".

** Use .247" top pin for non-control chambers.

Build-Up Pins	-9	-8	-7	-6	-5	-4	-3	-2	-1	0
	.037"	.051"	.066"	.080"	.093"	.107"	.120"	.135"	.149"	.163"
	+1	+2	+3	+4	+5	+6	+7	+8	+9	
	.177"	.192"	.205"	.218"	.232"	.247"	.261"	.275"	.289"	

.552" Diameter

	1	2	3	4	5	6	7	8	9	0
Bottom Pins	.213"	.228"	.241"	.256"	.269"	.283"	.297"	.311"	.326"	.340"
Master Pins		.028"	.042"	.056"	.070"	.084"	.098"	.112"	.126"	

* Standard top pin length is .171".

Build-Up Pins	-8	-7	-6	-5	-4	-3	-2	-1	0	
	.030"	.045"	.058"	.072"	.087"	.100"	.114"	.128"	.142"	
	+1	+2	+3	+4	+5	+6	+7	+8	+9	
	.156"	.171"	.184"	.198"	.212"	.226"	.241"	.253"	.268"	

Notes:

Corbin X Class (System 70)

46

MACS:	4	Key Origination	
Increment:	.028"	Blackhawk	
Progression:	Single Step	CodeMax	Varies
E.P.D.	Varies	HPC Blitz	CX5
Included Angle:	90°	HPC Punch	
Root Cut:	.053"	ITL	108

Root Depths	
1	.333"
2	.305"
3	.277"
4	.249"
5	.221"
6	.193"

Spacing	
1	.197"
2	.353"
3	.509"
4	.665"
5	.821"
6	.977"
7	1.133"

Notes
1) Read and cut bow-to-tip.
2) .509" diameter plugs have EPD of .504", .552" are .546"
3) For CodeMax, use DSD #301 for IC, DSD #3301 for Non-IC.
4) Use .320" top pin for PL5000 padlock / to block master ring.
5) 5 ½ or % cut used on construction master keys.

5 ½ or %	.207"
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T.F.C: .197"

B.C.C: .156"

.509" Diameter

	1	2	3	4	5	6				
Bottom Pins	.171"	.198"	.228"	.256"	.283"	.311"				
Master Pins	.028"	.056"	.084"	.122"	.140"					
IC Top Pins	.192"	.163"	.135"	.107"	.080"	.080"				

* Standard top pin length is .171".

** Use .247" top pin for non-control chambers.

	-4	-3	-2	-1	0	+1	+2	+3	+4	+5
Build-Up Pins	.051"	.080"	.107"	.135"	.163"	.192"	.218"	.247"	.275"	.303"

.552" Diameter

	1	2	3	4	5	6				
Bottom Pins	.213"	.241"	.269"	.297"	.326"	.354"				
Master Pins	.028"	.056"	.084"	.122"	.140"					

* Standard top pin length is .171".

	-4	-3	-2	-1	0	+1	+2	+3	+4	+5
Build-Up Pins	.030"	.058"	.087"	.114"	.142"	.171"	.198"	.226"	.253"	.282"

Notes:

Corbin Z Class (Pre-System 70)

47

MACS:	8	Key Origination		Root Depths		Spacing	
Increment:	.014"	Blackhawk	Card 438	1	.339"	1	.250"
Progression:	Two Step	CodeMax	Varies	2	.325"	2	.406"
E.P.D.	Varies	HPC Blitz	C14	3	.311"	3	.562"
Included Angle:	90°	HPC Punch		4	.297"	4	.718"
Root Cut:	.053"	ITL	113	5	.283"	5	.874"
Notes 1) Read and cut tip-to-bow. 2) .509" diameter plugs have EPD of .504", .552" are .546" 3) For CodeMax, use DSD #18 for IC, DSD #3018 for Non-IC. 4) Use .320" top pin for PL5000 padlock / to block master ring. 5) 11 cut used on some construction master keys.				6	.269"	6	1.030"
				7	.255"	7	1.186"
				8	.241"	T.F.C: .250"	
				9	.227"	B.C.C: .156"	
				10	.213"		
				11*	.199"		

.509" Diameter

	1	2	3	4	5	6	7	8	9	0
Bottom Pins	.160"	.175"	.189"	.203"	.217"	.231"	.245"	.260"	.273"	.288"
Master Pins		.028"	.042"	.056"	.070"	.084"	.098"	.112"	.126"	
IC Top Pins	.192"	.177"	.163"	.149"	.135"	.120"	.107"	.093"	.080"	.066"

* Standard top pin length is .171".

** Use .247" top pin for non-control chambers.

Build-Up Pins	-9	-8	-7	-6	-5	-4	-3	-2	-1	0
	.037"	.051"	.066"	.080"	.093"	.107"	.120"	.135"	.149"	.163"
	+1	+2	+3	+4	+5	+6	+7	+8	+9	
	.177"	.192"	.205"	.218"	.232"	.247"	.261"	.275"	.289"	

.552" Diameter

	1	2	3	4	5	6	7	8	9	0
Bottom Pins	.203"	.217"	.231"	.245"	.260"	.273"	.288"	.301"	.316"	.330"
Master Pins		.028"	.042"	.056"	.070"	.084"	.098"	.112"	.126"	

* Standard top pin length is .171".

Build-Up Pins	-8	-7	-6	-5	-4	-3	-2	-1	0	
	.030"	.045"	.058"	.072"	.087"	.100"	.114"	.128"	.142"	
	+1	+2	+3	+4	+5	+6	+7	+8	+9	
	.156"	.171"	.184"	.198"	.212"	.226"	.241"	.253"	.268"	

Notes:

Dexter (Post-1969)

48

MACS: 6
Increment: .015"
Progression: Two-Step
E.P.D. .485"
Included Angle: 100°
Root Cut: .046"

Key Origination	
Blackhawk	Card 409
CodeMax	DSD #25
HPC Blitz	C16
HPC Punch	P16
ITL	142

Root Depths	
0	.320"
1	.305"
2	.290"
3	.275"
4	.260"
5	.245"
6	.230"
7	.215"
8	.200"
9	.185"

Spacing	
1	.216"
2	.371"
3	.526"
4	.681"
5	.836"
6	.991"

T.F.C: .216"
B.C.C: .155"

Notes
1) Read and cut bow-to-tip
2) Blackhawk Card 409 lists depths 1-0, not 0-9.
3) Applies to Dexter large pin cylinders manufactured from 1969 on.

	0	1	2	3	4	5	6	7	8	9
Bottom Pins	.165"	.180"	.195"	.210"	.225"	.240"	.255"	.270"	.285"	.300"
Master Pins	-	-	.030"	.045"	.060"	.075"	.090"	.105"	.120"	.135"
Top Pins	.155"				.155"			.155"		

Notes:

Dexter (Pre-1969)

49

MACS: 4
Increment: .020"
Progression: Two Step
E.P.D. .487"
Included Angle: 100°
Root Cut: .050"

Key Origination	
Blackhawk	
CodeMax	DSD #24
HPC Blitz	
HPC Punch	
ITL	141

Root Depths	
0	.325"
1	.305"
2	.285"
3	.265"
4	.245"
5	.225"
6	.205"
7	.185"

Spacing	
1	.216"
2	.371"
3	.526"
4	.681"
5	.836"
6	.991"

T.F.C: .216"

B.C.C: .155"

Notes
1) Read and cut bow-to-tip
2) Applies to Dexter large pin cylinders manufactured prio to 1969.

	0	1	2	3	4	5	6	7
Bottom Pins	.162"	.182"	.202"	.222"	.242"	.262"	.282"	.302"
Master Pins			.040"	.060"	.080"	.100"	.120"	.140"
Top Pins	.180"							

Notes:

Dorma Large Pin (SKC)

50

MACS: 9
Increment: .0125"
Progression: Two Step
E.P.D. See Notes
Included Angle: 90°
Root Cut: .048"

Key Origination	
Blackhawk	
CodeMax	DSD #388
HPC Blitz	
HPC Punch	
ITL	97

Root Depths	
0	.318"
1	.3055"
2	.293"
3	.2805"
4	.268"
5	.2555"
6	.243"
7	.2305"
8	.218"
9	.2055"

Spacing	
1	.216"
2	.356"
3	.496"
4	.636"
5	.776"
6	.916"
7	1.056"

T.F.C: .216"
B.C.C: .140"

Notes
1) Read and cut tip-to-bow.
2) E.P.D for SFIC is .428", everything else is .503".

SFIC Bottom Pins	0	1	2	3	4	5	6	7	8	9
	.110"	.1225"	.135"	.1475"	.160"	.1725"	.185"	.1975"	.210"	.2225"

Full Size Bottom Pins	0	1	2	3	4	5	6	7	8	9
	.185"	.1975"	.210"	.2225"	.235"	.2475"	.260"	.2725"	.285"	.2975"

Master, Build-Up, & Top Pins	2	3	4	5	6	7	8	9	10	11
	.025"	.0375"	.050"	.0625"	.075"	.0875"	.100"	.1125"	.125"	.1375"
	12	13	14	15	16	17	18	19		
	.150"	.1625"	.175"	.1875"	.200"	.2125"	.225"	.2375"		

Fixed Cylinder Top Pins*	A	B	C	D						
	.200"	.235"	.285"	.2975"						

*A is used with depths 5-9 in KIK cylinders, B is used with depths 0-4 in KIK cylinders, C is used with depths 5-9 in rim & mortise cylinders, D is used with depths 0-4 in rim & mortise cylinders.

Notes:

Falcon .025"

51

MACS: 5
Increment: .025"
Progression: Single Step
E.P.D. .504"
Included Angle: 90°
Root Cut: .040"

Key Origination	
Blackhawk	
CodeMax	DSD #625
HPC Blitz	
HPC Punch	
ITL	

Root Depths	
0 (1)	.339"
1 (2)	.314"
2 (3)	.289"
3 (4)	.264"
4 (5)	.239"
5 (6)	.214"
6 (7)	.189"

Spacing	
1	.237"
2	.393"
3	.549"
4	.705"
5	.861"
6	1.017"
7	1.173"

Notes
1) Read and cut bow-to-tip.
2) Used for Falcon "Restricted" keyway cylinders.
3) Falcon numbers depths 0-6, HPC numbers depths 1-7.

T.F.C: .237"

B.C.C: .156"

	0	1	2	3	4	5	6			
Bottom Pins	.165"	.190"	.215"	.240"	.265"	.290"	.315"			
Master Pins	.025"	.050"	.075"	.100"	.125"	.150"				
Top Pins	.185"	.185"	.185"	.185"	.185"	.185"	.185"			

Notes:

MACS: 9
Increment: .0125"
Progression: Two Step
E.P.D. See Notes
Included Angle: 90°
Root Cut: .048"

Key Origination	
Blackhawk	
CodeMax	
HPC Blitz	*
HPC Punch	
ITL	*

Root Depths	
0	.318"
1	.3055"
2	.293"
3	.2805"
4	.268"
5	.2555"
6	.243"
7	.2305"
8	.218"
9	.2055"

Spacing	
1	.216"
2	.356"
3	.496"
4	.637"
5	.777"
6	.917"
7	1.057"

T.F.C: .216"

B.C.C: .140"

Notes
1) Read and cut bow-to-tip.
2) For non-IC use preloaded top pins, place bottom and master pins in plug as needed.
3) Use Blitz cards GMS CARD1-MXA or C1215, GMS provides programming guide for ITL 9700.
4) SFIC EPD: .428", all others .499".

Bottom Pins (SFIC)	0A	1A	2A	3A	4A	5A	6A	7A	8A	9A
	.110"	.1225"	.135"	.1475"	.160"	.1725"	.185"	.1975"	.210"	.2225"

Master, Build-Up, & Top Pins (SFIC)	2B	3B	4B	5B	6B	7B	8B	9B	10B	11B
	.025"	.0375"	.050"	.0625"	.075"	.0875"	.100"	.1125"	.125"	.1375"
	12B	13B	14B	15B	16B	17B	18B	19B		
	.150"	.1625"	.175"	.1875"	.200"	.2125"	.225"	.2375"		

Bottom Pins (Fixed Cyl. & LFIC/FSIC)	0	1	2	3	4	5	6	7	8	9
	.181"	.1935"	.206"	.2185"	.231"	.2435"	.256"	.2685"	.281"	.2935"

Master & Top Pins (Fixed Cyl. & LFIC/FSIC)	2	3	4	5	6	7	8	9		
	.025"	.0375"	.050"	.0625"	.075"	.0875"	.100"	.1125"		
	Top, Long		Top, Short							
	.217"		.197"							

Notes:

MACS: 7
Increment: .015"
Progression: Two Step
E.P.D. .500"
Included Angle: 100°
Root Cut: .031"

Key Origination	
Blackhawk	Card 198
CodeMax	DSD #60
HPC Blitz	C45
HPC Punch	P45
ITL	391

Root Depths	
0	.335"
1	.320"
2	.305"
3	.290"
4	.275"
5	.260"
6	.245"
7	.230"
8	.215"
9	.200"

Spacing	
1	.231"
2	.387"
3	.543"
4	.699"
5	.855"
6	1.011"

T.F.C: .231"

B.C.C: .156"

Notes
1) Read and cut tip-to-bow.
2) Cut on restricted blanks.

	0	1	2	3	4	5	6	7	8	9
Bottom Pins	.165"	.180"	.195"	.210"	.225"	.240"	.255"	.270"	.285"	.300"
Master Pins			.030"	.045"	.060"	.075"	.090"	.105"	.120"	
Top Pins	.235"				.200"			.165"		

Notes:

Hager H Series

54

MACS:	Varies
Increment:	.0125"
Progression:	Two Step
E.P.D.	Varies
Included Angle:	90°
Root Cut:	.048"

Key Origination	
Blackhawk	
CodeMax	
HPC Blitz	
HPC Punch	
ITL	

Root Depths	
0	.318"
1	.3055"
2	.293"
3	.2805"
4	.268"
5	.2555"
6	.243"
7	.2305"
8	.218"
9	.2055"

Spacing	
1	.2158"
2	.3559"
3	.4961"
4	.6362"
5	.7764"
6	.9166"
7	1.0567"

T.F.C: .2158"
B.C.C: .140"

Notes
1) Read and cut bow-to-tip.
2) Manufacturer recommends MACS of 8 but 9 is possible.
3) KIK/KIL top pins: Plug total 0-4 = .235", 5-9 = .200".
4) Rim/mortise top pins: Plug total = 0-4 = .305", 5-9 = .280".
5) Conventional cylinder springs: LAB 108BV1.

Conventional Cylinder Pins (.108" Pin Diameter)

	0	1	2	3	4	5	6	7	8	9
Bottom Pins	.1850"	.1975"	.2100"	.2225"	.2350"	.2475"	.2600"	.2725"	.2850"	.2975"

SFIC Pins

	0A	1A	2A	3A	4A	5A	6A	7A	8A	9A
Bottom Pins	.110"	.123"	.135"	.148"	.160"	.173"	.185"	.198"	.210"	.223"

Master, Build-Up, & Top Pins (All Cylinders)	2B	3B	4B	5B	6B	7B	8B	9B	10B	11B
	.025"	.0375"	.050"	.0625"	.075"	.0875"	.100"	.1125"	.125"	.1375"
	12B	13B	14B	15B	16B	17B	18B	19B		
	.150"	.1625"	.175"	.1875"	.200"	.2125"	.225"	.2375"		

Notes:

Huber Large Pin

55

MACS: 7
Increment: .014"
Progression: Two Step
E.P.D. .508"
Included Angle: 100°
Root Cut: .046"

Key Origination	
Blackhawk	
CodeMax	DSD #330/331
HPC Blitz	
HPC Punch	
ITL	216

Root Depths	
0	.322"
1	.308"
2	.294"
3	.280"
4	.266"
5	.252"
6	.238"
7	.224"
8	.210"
9	.196"

Spacing	
1	.195"
2	.360"
3	.525"
4	.690"
5	.855"

T.F.C: .195"
B.C.C: .165"

Notes
1) Read and cut-bow-to tip.
2) Uses Ilco 1020/Curtis HU-1/Star 5HU1 blanks.
3) Alternate spacing of .195", .351", .507", .663", .819" has been reported in some sources, use DSD #330 for this and reduce MACS to 6.

	0	1	2	3	4	5	6	7	8	9
Bottom Pins	.186"	.200"	.214"	.228"	.242"	.256"	.270"	.284"	.298"	.312"
Top Pins	.168"									

Notes:

Ilco Small Pin

MACS: 4
Increment: .018"
Progression: N/A
E.P.D. .398"
Included Angle: 90°
Root Cut: .044"

Key Origination	
Blackhawk	Card 186
CodeMax	DSD #41
HPC Blitz	C27
HPC Punch	
ITL	235

Root Depths	
1	.270"
2	.252"
3	.234"
4	.216"
5	.198"
6	.180"
7	.162"

Spacing	
1	.162"
2	.287"
3	.412"
4	.537"
5	.662"

Notes
1) Read and cut bow-to-tip. 2) Most examples use .100" diameter pins. 3) Top pin sizes varied by cylinder format, the sizes listed here are for peanut cylinders and small padlocks.

T.F.C: .162"
B.C.C: .125"

	1	2	3	4	5	6	7			
Bottom Pins	.128"	.146"	.164"	.182"	.200"	.218"	.236"			
Top Pins	.148"			.112"			.094"			

Notes:

Kwikset .023"

57

MACS: 4
Increment: .023"
Progression: Single Step
E.P.D. .501"
Included Angle: 90°
Root Cut: .084"

Key Origination	
Blackhawk	Card 407
CodeMax	DSD #46
HPC Blitz	C31
HPC Punch	P31
ITL	265

Root Depths	
1	.329"
2	.306"
3	.283"
4	.260"
5	.237"
6	.214"
7	.191"

Spacing	
1	.247"
2	.397"
3	.547"
4	.697"
5	.847"

T.F.C: .247"

B.C.C: .150"

Notes
1) Read and cut bow-to-tip.
2) Use .160" length top pins for construction and master keying.
3) Used in Corbin 80/Ruswin 751 residential keyways between 1966 - 1992 under "K Class" name.
5) 7th depth used in factory generated master key systems, not usable in SmartKey. No factory #7 pins exist.
6) Factory plugs are .498" diameter but bevels in pins and plug allow a ±.007" variance in EPD (.491"-.505").

	1	2	3	4	5	6	7			
Bottom Pins	.172"	.195"	.218"	.241"	.264"	.287"				
Master Pins	.023"	.046"	.069"	.092"	.115"	.138"				
Top Pins	.180"						.160"			

Notes:

Kwikset Titan

58

MACS: 4
Increment: .023"
Progression: Single Step
E.P.D. .498"
Included Angle: 90°
Root Cut: .084"

Key Origination	
Blackhawk	
CodeMax	DSD #94
HPC Blitz	C31X
HPC Punch	P31X
ITL	250

Root Depths	
1	.329"
2	.306"
3	.283"
4	.260"
5	.237"
6	.214"
7	.191"

Spacing	
1	.097"
2	.247"
3	.397"
4	.547"
5	.697"
6	.847"

T.F.C: .097"

B.C.C: .150"

Notes
1) Read and cut bow-to-tip.
2) Use .160" length top pins for construction keying.
3) 7th depth used in factory generated, non-SmartKey systems.
4) Rekeying notch is cut on the back edge of the key. The notch is .249" deep, with a flat .188" wide starting .228" from the shoulder and extending towards the tip.
5) When using Kwikset original (flat nose) pins in position 1, the cut flat must be extended fully to the shoulder.

	1	2	3	4	5	6	7			
Bottom Pins	.172"	.195"	.218"	.241"	.264"	.287"	.310"			
Master Pins	.023"	.046"	.069"	.092"	.115"	.138"				
Top Pins	.180"									

*Titan uses the same pins as Kwikset Classic. Spool top pins are available in both .180" and .160" lengths but are rarely used in factory combined locks.

Notes:

Lockwood Large Pin (Class 1004)

59

MACS: 6
Increment: .015"
Progression: Two Step
E.P.D. .498"
Included Angle: 100°
Root Cut: .046"

Key Origination	
Blackhawk	
CodeMax	DSD #356
HPC Blitz	CMK33
HPC Punch	
ITL	

Root Depths	
0	.335"
1	.320"
2	.305"
3	.290"
4	.275"
5	.260"
6	.245"
7	.230"
8	.215"
9	.200"
10	.185"
11	.170"

Spacing	
1	.277"
2	.433"
3	.589"
4	.745"
5	.901"
6	1.057"
7	1.213"

T.F.C: .240"

B.C.C: .156"

Notes
1) Read and cut bow-to-tip.
2) Many locks may only use depths 0-9

	0	1	2	3	4	5	6	7	8	9	10	11
Bottom Pins	.163"	.178"	.193"	.208"	.223"	.238"	.253"	.268"	.283"	.298"	.313"	.328"
Master Pins			.030"	.045"	.060"	.075"	.090"	.105"	.120"	.135"	.150"	.165"
Top Pins	.235"			.220"			.200"			.165"		

Notes:

Lockwood Large Pin (Standard)

60

MACS: 5
Increment: .018"
Progression: Two Step
E.P.D. .499"
Included Angle: 100°
Root Cut: .080"

Key Origination	
Blackhawk	Card 184
CodeMax	DSD #43
HPC Blitz	C28
HPC Punch	C28
ITL	266

Root Depths	
0	.320"
1	.302"
2	.284"
3	.266"
4	.248"
5	.230"
6	.212"
7	.194"
8	.176"
9	.158"

Spacing	
1	.277"
2	.433"
3	.589"
4	.745"
5	.901"
6	1.057"

T.F.C: .277"
B.C.C: .156"

Notes
1) Read and cut bow-to-tip.
2) .185" top pins were used in older LFIC cylinders.

	0	1	2	3	4	5	6	7	8	9
Bottom Pins	.179"	.197"	.215"	.233"	.251"	.269"	.287"	.305"	.323"	.341"
Master Pins			.036"	.054"	.072"	.090"	.108"	.126"	.144"	.162"
Top Pins	.162"									

Notes:

LSDA - LSA Keyway

61

MACS: 6
Increment: .018"
Progression: Two Step
E.P.D. .502"
Included Angle: 100°
Root Cut: .055"

Key Origination	
Blackhawk	
CodeMax	DSD #605
HPC Blitz	C101
HPC Punch	
ITL	254

Root Depths	
0	.322"
1	.304"
2	.286"
3	.268"
4	.250"
5	.232"
6	.214"
7	.196"
8	.178"
9	.160"

Spacing	
1	.216"
2	.372"
3	.528"
4	.684"
5	.840"
6	.996"

T.F.C: .216"
B.C.C: .156"

Notes
1) Read and cut bow-to-tip.
2) Cylinders use .220" top pin for pin stacks of 0 - 8, pin stacks of 9 use .1605" top pins.
3) Reverse of Reading "HA" keyway.

	0	1	2	3	4	5	6	7	8	9
Bottom Pins	.180"	.198"	.216"	.234"	.252"	.270"	.288"	.306"	.324"	.342"
Master Pins			.036"	.054"	.072"	.090"	.108"	.126"	.144"	.162"
Top Pins	See Notes.									

Notes:

Master Lock Large Pin (1176 Keyway)

62

MACS: 4
Increment: .023"
Progression: Single Step
E.P.D. .498"
Included Angle: 90°
Root Cut: .084"

Key Origination	
Blackhawk	Card 407
CodeMax	DSD #46
HPC Blitz	C31
HPC Punch	P31
ITL	265

Root Depths	
1	.329"
2	.306"
3	.283"
4	.260"
5	.237"
6	.214"
7	.191"

Spacing	
1	.247"
2	.397"
3	.547"
4	.697"
5	.847"

T.F.C: .247"

B.C.C: .150"

Notes
1) Read and cut bow-to-tip. 2) Identified as 1176 Keyway, for 1145 Keyway products refer to Schlage Large Pin. 3) When using BumpStop driver pins in 1145 Keyway products they must be paired with a spool bottom pin. 4) BumpStop driver pins must be paired with a bottom pin of one or the 3 shortest sizes (1, 2, or 3 in 1176 keyway).

	1	2	3	4	5	6	7			
Bottom Pins*	.172"	.195"	.218"	.241"	.264"	.287"	.310"			
Master Pins	.023"	.046"	.069"	.092"	.115"	.138"				
Top Pins	.180"									

* Bottom pins available in both standard and spooled types.

Notes:

Master Lock Pro Series

MACS: 5
Increment: .0155"
Progression: Two Step
E.P.D. .397"
Included Angle: 90°
Root Cut: .044"

Key Origination	
Blackhawk	Card 327
CodeMax	DSD #97
HPC Blitz	C103A
HPC Punch	
ITL	249

Root Depths		
	Pre 2001	Post 2001
0	.2710"	.2845"
1	.2555"	.2690"
2	.2400"	.2535"
3	.2245"	.2380"
4	.2090"	.2225"
5	.1935"	.2070"
6	.1780"	.1915"
7	.1625"	.1760"

Spacing	
1	.187"
2	.312"
3	.437"
4	.562"
5	.687"
6	.812"
7	.937"

Notes
1) Read and cut bow-to-tip.
2) For use with blanks 6000K and 7000K.
3) For pre-2001, 'flat back' blanks use DSD #50, C103, or ITL 287.

T.F.C: .187"
B.C.C: .125"

	0	1	2	3	4	5	6	7		
Bottom Pins	.127"	.142"	.1575"	.173"	.1885"	.204"	.2195"	.235"		
Master Pins		.0155"	.031"	.0465"	.062"	.0775"	.093"	.1085"		
Top Pins*	.134"									

*Top pins are available as serrated or spool.

Notes:

Norwalk Large Pin (.025")

64

MACS: N/A

Increment: .025"

Progression: Two Step

E.P.D. .557"

Included Angle: 90°

Root Cut: .044"

Key Origination	
Blackhawk	
CodeMax	
HPC Blitz	
HPC Punch	
ITL	

Root Depths	
1	.320"
2	.295"
3	.270"
4	.245"
5	.220"

Spacing	
1	.210"
2	.385"
3	.560"
4	.735"
5	.910"

Notes
1) Read and cut-bow-to tip.
2) Top pins were roughly machined and often had spurs or only partially cut surfaces facing the springs.

T.F.C: .210"

B.C.C: .175"

	1	2	3	4	5					
Bottom Pins	.237"	.262"	.237"	.212"	.187"					
Master Pins	.025"	.050"	.075"	.100"						
Top Pins	~.187"									

Notes:

Old Russwin 752 Class

65

MACS: 5
Increment: .020"
Progression: Single Step
E.P.D. Varies
Included Angle: 90°
Root Cut: .053"

Key Origination	
Blackhawk	
CodeMax	Varies
HPC Blitz	CX13
HPC Punch	
ITL	

Root Depths	
0	.310"
1	.290"
2	.270"
3	.250"
4	.230"
5	.210"
6	.190"

Spacing	
1	.250"
2	.406"
3	.562"
4	.718"
5	.874"

T.F.C.: .250"

B.C.C.: .156"

Notes
1) Read and cut bow-to-tip.
2) .509" diameter plugs have EPD of .496", .552" are .541"
3) For CodeMax, use DSD #324 for IC, DSD #3324 for Non-IC.
4) Use .320" top pin for PL5000 padlock and block master ring.

.509" Diameter

	0	1	2	3	4	5	6			
Bottom Pins	.186"	.208"	.228"	.248"	.267"	.288"	.307"			
Master Pins			.040"	.060"	.080"	.100"	.120"			
IC Top Pins	.184"	.163"	.142"	.122"	.103"	.082"	.063"			

* Standard top pin length is .171".

** Use .247" top pin for non-control chambers.

Build-Up Pins	-6	-5	-4	-3	-2	-1	0			
	.042"	.063"	.082"	.103"	.122"	.142"	.163"			
	+1	+2	+3	+4	+5	+6				
	.184"	.202"	.222"	.241"	.261"	.282"				

.552" Diameter

	0	1	2	3	4	5	6			
Bottom Pins	.231"	.251"	.269"	.291"	.311"	.330"	.349"			
Master Pins			.040"	.060"	.080"	.100"	.120"			

* Standard top pin length is .171".

Build-Up Pins	-5	-4	-3	-2	-1	0				
	.042"	.063"	.082"	.103"	.122"	.142"				
	+1	+2	+3	+4	+5	+6				
	.163"	.184"	.202"	.222"	.241"	.261"				

Notes:

Old Russwin 852 Class

66

MACS: 5
Increment: .020"
Progression: Single Step
E.P.D. Varies
Included Angle: 90°
Root Cut: .053"

Key Origination	
Blackhawk	Card 427
CodeMax	Varies
HPC Blitz	C852
HPC Punch	
ITL	

Root Depths	
B	.325"
0	.305"
1	.285"
2	.265"
3	.245"
4	.225"
5	.205"
6	.185"

Spacing	
1	.250"
2	.406"
3	.562"
4	.718"
5	.874"
6	1.030"

T.F.C.: .250"

B.C.C.: .156"

Notes

- 1) Read and cut bow-to-tip.
- 2) .509" diameter plugs have EPD of .496", .552" are .542"
- 3) For CodeMax, use DSD #507 for IC, DSD #3507 for Non-IC.
- 4) Use .320" top pin for PL5000 padlock and block master ring.

.509" Diameter

	B	0	1	2	3	4	5	6	7	
Bottom Pins	.171"	.193"	.213"	.231"	.251"	.273"	.291"	.311"		
Master Pins				.040"	.060"	.080"	.100"	.120"	.140"	
IC Top Pins	.202"	.184"	.163"	.142"	.122"	.103"	.082"	.063"		

* Standard top pin length is .171".

** Use .247" top pin for non-control chambers.

	-6	-5	-4	-3	-2	-1	0			
Build-Up Pins	.042"	.063"	.082"	.103"	.122"	.142"	.163"			
	+1	+2	+3	+4	+5	+6				
	.184"	.202"	.222"	.241"	.261"	.282"				

.552" Diameter

	B	0	1	2	3	4	5	6	7	
Bottom Pins	.217"	.234"	.256"	.276"	.294"	.316"	.334"	.354"		
Master Pins				.040"	.060"	.080"	.100"	.120"	.140"	

* Standard top pin length is .171".

	-5	-4	-3	-2	-1	0				
Build-Up Pins	.042"	.063"	.082"	.103"	.122"	.142"				
	+1	+2	+3	+4	+5	+6				
	.163"	.184"	.202"	.222"	.241"	.261"				

Notes:

Old Russwin A Class

67

MACS: 5
Increment: .020"
Progression: Single Step
E.P.D. Varies
Included Angle: 90°
Root Cut: .053"

Key Origination	
Blackhawk	Card 440
CodeMax	Varies
HPC Blitz	CX12
HPC Punch	
ITL	377

Root Depths	
0	.315"
1	.295"
2	.275"
3	.255"
4	.235"
5	.215"
6	.195"

Spacing	
1	.250"
2	.406"
3	.562"
4	.718"
5	.874"
6	1.030"

T.F.C: .250"
B.C.C: .156"

Notes
1) Read and cut bow-to-tip.
2) .509" diameter plugs have EPD of .492", .552" are .535"
3) For CodeMax, use DSD #80 for IC, DSD #3080 for Non-IC.
4) Use .320" top pin for PL5000 padlock and block master ring.

.509" Diameter

	0	1	2	3	4	5	6			
Bottom Pins	.177"	.198"	.217"	.236"	.256"	.276"	.297"			
Master Pins			.040"	.060"	.080"	.100"	.120"			
IC Top Pins	.184"	.163"	.142"	.122"	.103"	.082"	.063"			

* Standard top pin length is .171".

** Use .247" top pin for non-control chambers.

	-6	-5	-4	-3	-2	-1	0			
Build-Up Pins	.042"	.063"	.082"	.103"	.122"	.142"	.163"			
	+1	+2	+3	+4	+5	+6				
	.184"	.202"	.222"	.241"	.261"	.282"				

.552" Diameter

	0	1	2	3	4	5	6			
Bottom Pins	.220"	.241"	.260"	.279"	.301"	.320"	.340"			
Master Pins			.040"	.060"	.080"	.100"	.120"			

* Standard top pin length is .171".

	-5	-4	-3	-2	-1	0				
Build-Up Pins	.042"	.063"	.082"	.103"	.122"	.0142"				
	+1	+2	+3	+4	+5	+6				
	.163"	.184"	.202"	.222"	.241"	.261"				

Notes:

Old Russwin Small Pin (Auto Cylinders)

68

MACS: 3
Increment: .020"
Progression: Two Step
E.P.D. .390"
Included Angle: 90°
Root Cut: .0625"

Key Origination	
Blackhawk	Card 429
CodeMax	
HPC Blitz	
HPC Punch	
ITL	

Root Depths	
0	.265"
1	.245"
2	.225"
3	.205"
4	.185"
5	.165"
6	.145"

Spacing	
1	.156"
2	.281"
3	.406"
4	.531"

T.F.C: .156"

B.C.C: .125"

Notes
1) Read and cut bow-to-tip.
2) Some other auto cylinders us spacing of .156"/.140" on longer blank.

	0	1	2	3	4	5	6			
Bottom Pins	.125"	.145"	.165"	.185"	.205"	.225"	.245"			
Master Pins			.040"	.060"	.080"	.100"	.120"			
Top Pins	.070"	.198"	.184"	.170"	.156"	.142"	.126"			

Notes:

Old Russwin Small Pin (Rim Cylinders)

69

MACS: 5
Increment: .020"
Progression: Two Step
E.P.D. .390"
Included Angle: 90°
Root Cut: .0625"

Key Origination	
Blackhawk	Card 462
CodeMax	DSD #340
HPC Blitz	
HPC Punch	
ITL	

Root Depths	
0	.265"
1	.245"
2	.225"
3	.205"
4	.185"
5	.165"
6	.145"
7	.125"

Spacing	
1	.254"
2	.394"
3	.534"
4	.674"

T.F.C: .254"

B.C.C: .140"

Notes
1) Read and cut bow-to-tip.

	0	1	2	3	4	5	6			
Bottom Pins	.125"	.145"	.165"	.185"	.205"	.225"	.245"			
Master Pins			.040"	.060"	.080"	.100"	.120"			
Top Pins	.070"	.198"	.184"	.170"	.156"	.142"	.126"			

Notes:

Russwin 981 Class

70

MACS: 7
Increment: .015"
Progression: Two Step
E.P.D. Varies
Included Angle: 90°
Root Cut: .053"

Key Origination	
Blackhawk	Card 426
CodeMax	Varies
HPC Blitz	C981
HPC Punch	
ITL	379

Root Depths	
B	.338"
0	.323"
1	.308"
2	.293"
3	.278"
4	.263"
5	.248"
6	.233"
7	.218"
8	.203"
9*	.188"

Spacing	
1	.250"
2	.406"
3	.562"
4	.718"
5	.874"
6	1.030"
7	1.186"

T.F.C: .250"

B.C.C: .156"

Notes
1) Read and cut bow-to-tip.
2) .509" diameter plugs have EPD of .498", .552" are .541"
3) For CodeMax, use DSD 508 for IC, DSD 3508 for Non-IC.
4) Use .320" top pin for PL5000 padlock and block master ring.

.509" Diameter

	B	0	1	2	3	4	5	6	7	8	9
Bottom Pins	.160"	.175"	.189"	.203"	.220"	.234"	.248"	.263"	.279"	.294"	
Master Pins				.030"	.045"	.060"	.075"	.090"	.105"	.120"	.135"
IC Top Pins	.192"	.177"	.163"	.149"	.133"	.118"	.103"	.087"	.072"	.058"	

* Standard top pin length is .171".

** Use .247" top pin for non-control chambers.

Build-Up Pins	-9	-8	-7	-6	-5	-4	-3	-2	-1	0
	.028"	.042"	.058"	.072"	.087"	.103"	.118"	.133"	.149"	.163"
	+1	+2	+3	+4	+5	+6	+7	+8	+9	
	.177"	.192"	.208"	.222"	.238"	.253"	.268"	.282"	.298"	

.552" Diameter

	B	0	1	2	3	4	5	6	7	8	9
Bottom Pins	.203"	.217"	.231"	.248"	.263"	.276"	.291"	.307"	.322"	.337"	
Master Pins				.030"	.045"	.060"	.075"	.090"	.105"	.120"	.135"

* Standard top pin length is .171".

Build-Up Pins	-7	-6	-5	-4	-3	-2	-1	0		
	.037"	.051"	.066"	.082"	.098"	.112"	.126"	.142"		
	+1	+2	+3	+4	+5	+6	+7	+8	+9	
	.156"	.171"	.187"	.202"	.218"	.232"	.247"	.261"	.277"	

Notes:

Russwin DH Class (Pre-System 70)

71

MACS: 7
Increment: .015"
Progression: Two Step
E.P.D. Varies
Included Angle: 90°
Root Cut: .053"

Key Origination	
Blackhawk	Card 428
CodeMax	Varies
HPC Blitz	C42
HPC Punch	
ITL	379

Root Depths	
0	.341"
1	.326"
2	.311"
3	.296"
4	.281"
5	.266"
6	.251"
7	.236"
8	.221"
9	.206"

Spacing	
1	.250"
2	.406"
3	.562"
4	.718"
5	.874"
6	1.030"
7	1.186"

T.F.C: .250"

B.C.C: .156"

10*	.191"
-----	-------

Notes
1) Read and cut bow-to-tip.
2) .509" diameter plugs have EPD of .501", .552" are .544"
3) For CodeMax, use DSD #57 for IC, DSD #3057 for Non-IC.
4) Use .320" top pin for PL5000 padlock / to block master ring.
5) 10 cut used on some construction master keys.

.509" Diameter

	0	1	2	3	4	5	6	7	8	9
Bottom Pins	.160"	.175"	.189"	.203"	.220"	.234"	.248"	.263"	.279"	.294"
Master Pins			.030"	.045"	.060"	.075"	.090"	.105"	.120"	.135"
IC Top Pins	.192"	.177"	.163"	.149"	.133"	.118"	.103"	.087"	.072"	.058"

* Standard top pin length is .171".

** Use .247" top pin for non-control chambers.

Build-Up Pins	-9	-8	-7	-6	-5	-4	-3	-2	-1	0
	.028"	.042"	.058"	.072"	.087"	.103"	.118"	.133"	.149"	.163"
	+1	+2	+3	+4	+5	+6	+7	+8	+9	
	.177"	.192"	.208"	.222"	.238"	.253"	.263"	.282"	.298"	

.552" Diameter

	0	1	2	3	4	5	6	7	8	9
Bottom Pins	.203"	.217"	.231"	.248"	.263"	.276"	.291"	.307"	.322"	.337"
Master Pins			.030"	.045"	.060"	.075"	.090"	.105"	.120"	.135"

* Standard top pin length is .171".

Build-Up Pins	-7	-6	-5	-4	-3	-2	-1	0	+1	+2
	.037"	.051"	.066"	.082"	.098"	.112"	.126"	.142"	.156"	.171"
	+3	+4	+5	+6	+7	+8	+9			
	.187"	.202"	.218"	.232"	.247"	.261"	.277"			

Notes:

Russwin N Class (Pre-System 70)

72

MACS: 7
Increment: .015"
Progression: Two Step
E.P.D. Varies
Included Angle: 90°
Root Cut: .053"

Key Origination	
Blackhawk	
CodeMax	Varies
HPC Blitz	C41
HPC Punch	
ITL	380

Root Depths	
0	.328"
1	.313"
2	.298"
3	.283"
4	.268"
5	.253"
6	.238"
7	.223"
8	.208"
9	.193"

Spacing	
1	.250"
2	.406"
3	.562"
4	.718"
5	.874"
6	1.030"
7	1.186"

T.F.C: .250"

B.C.C: .156"

Notes
1) Read and cut bow-to-tip
2) .509" diameter plugs have EPD of .499", .552" are .541"
3) For CodeMax, use DSD #56 for IC, DSD #3056 for Non-IC.
4) Use .320" top pin for PL5000 padlock / to block master ring.
5) 10 cut used on some constructions master keys.

10	.178"
----	-------

.509" Diameter

	0	1	2	3	4	5	6	7	8	9
Bottom Pins	.171"	.186"	.201"	.217"	.231"	.245"	.260"	.276"	.291"	.307"
Master Pins			.030"	.045"	.060"	.075"	.090"	.105"	.120"	.135"
IC Top Pins	.192"	.177"	.163"	.149"	.133"	.118"	.103"	.087"	.072"	.058"

* Standard top pin length is .171".

** Use .247" top pin for non-control chambers.

Build-Up Pins	-9	-8	-7	-6	-5	-4	-3	-2	-1	0
	.028"	.042"	.058"	.072"	.087"	.103"	.118"	.133"	.149"	.163"
	+1	+2	+3	+4	+5	+6	+7	+8	+9	
	.177"	.192"	.208"	.222"	.238"	.253"	.268"	.282"	.298"	

.552" Diameter

	0	1	2	3	4	5	6	7	8	9
Bottom Pins	.213"	.228"	.245"	.260"	.273"	.288"	.305"	.320"	.334"	.349"
Master Pins			.030"	.045"	.060"	.075"	.090"	.105"	.120"	.135"

* Standard top pin length is .171".

Build-Up Pins	-7	-6	-5	-4	-3	-2	-1	0		
	.037"	.051"	.066"	.082"	.098"	.112"	.126"	.142"		
	+1	+2	+3	+4	+5	+6	+7	+8	+9	
	.156"	.171"	.187"	.202"	.218"	.232"	.247"	.261"	.277"	

Notes:

Russwin N Class (System 70)

73

MACS: 4
Increment: .028"
Progression: Single Step
E.P.D. Varies
Included Angle: 90°
Root Cut: .053"

Key Origination	
Blackhawk	
CodeMax	Varies
HPC Blitz	CX7
HPC Punch	
ITL	378

Root Depths	
1	.334"
2	.306"
3	.278"
4	.250"
5	.222"
6	.194"

Spacing	
1	.250"
2	.406"
3	.562"
4	.718"
5	.874"
6	1.030"
7	1.186"

Notes
1) Read and cut bow-to-tip.
2) .509" diameter plugs have EPD of .499", .552" are .542"
3) For CodeMax, use DSD 77 for IC, DSD 3077 for Non-IC.
4) Use .320" top pin for PL5000 padlock and block master ring.

5 1/2	.208"
-------	-------

T.F.C: .250"

B.C.C: .156"

.509" Diameter

	1	2	3	4	5	6				
Bottom Pins	.165"	.193"	.220"	.248"	.276"	.305"				
Master Pins	.028"	.056"	.084"	.112"	.140"					
IC Top Pins	.192"	.163"	.135"	.107"	.080"	.080"				

* Standard top pin length is .171".

** Use .247" top pin for non-control chambers.

	-4	-3	-2	-1	0	+1	+2	+3	+4	+5
Build-Up Pins	.051"	.080"	.107"	.135"	.163"	.192"	.218"	.247"	.275"	.303"

.552" Diameter

	1	2	3	4	5	6				
Bottom Pins	.208"	.236"	.263"	.291"	.320"	.349"				
Master Pins	.028"	.056"	.084"	.112"	.140"					

* Standard top pin length is .171".

	-4	-3	-2	-1	0	+1	+2	+3	+4	+5
Build-Up Pins	.030"	.058"	.087"	.114"	.142"	.171"	.198"	.226"	.253"	.282"

Notes:

Sargent .020"

74

MACS: 7; Varies
Increment: .020"
Progression: Two-Step
E.P.D. .498"
Included Angle: 76°
Root Cut: .051"

Key Origination	
Blackhawk	Card 196
CodeMax	DSD #59
HPC Blitz	C44
HPC Punch	
ITL	388

Root Depths	
1	.330"
2	.310"
3	.290"
4	.270"
5	.250"
6	.230"
7	.210"
8	.190"
9	.170"
10	.150"

Spacing	
1	.215"
2	.371"
3	.527"
4	.683"
5	.839"
6	.995"

T.F.C: .215"

B.C.C: .156"

Notes
1) Read and cut bow-to-tip.
2) Used in all current Sargent cores/cylinders and in the following keyway families: A, B, C, G, H, K, L, N, R, V
3) MACS has been listed as 9 on older literature due to specialized cutting wheels no longer used. Follow code machines instructions for producing keys.

	1	2	3	4	5	6	7	8	9	10
Bottom Pins	.170"	.190"	.210"	.230"	.250"	.270"	.290"	.310"	.330"	.350"
Master Pins	—	.040"	.060"	.080"	.100"	.120"	.140"	.160"	.180"	.200"
Top Pins	.200"	.200"	.200"	.200"	.200"	.200"	.200"	.200"	.200"	.200"

6300 Series

	11	12	13	14						
Master Pins	.220"	.240"	.260"	.280"						

Notes:

Sargent Signature Series

75

MACS: 7
Increment: .020"
Progression: Two-Step
E.P.D. .498"
Included Angle: 79°
Root Cut: .051"

Key Origination	
Blackhawk	Card 196
CodeMax	DSD #59
HPC Blitz	C44
HPC Punch	
ITL	388

Root Depths	
1	.330"
2	.310"
3	.290"
4	.270"
5	.250"
6	.230"
7	.210"
8	.190"
9	.170"
10	.150"

Spacing	
1	.215"
2	.371"
3	.527"
4	.683"
5	.839"
6	.995"

T.F.C: .215"

B.C.C: .156"

Notes
1) Read and cut bow-to-tip.
2) Older listings provide a MACS of 9 using a radiused cut and pins.
3) Side pins (x6) and blocking pins (x2) are loaded into the cylinder at the factory and are not considered field servicable.
4) Signature system available in most Sargent keyway families.

	1	2	3	4	5	6	7	8	9	10
Bottom Pins	.170"	.190"	.210"	.230"	.250"	.270"	.290"	.310"	.330"	.350"
Master/Top Pins*	—	.040"	.060"	.080"	.100"	.120"	.140"	.160"	.180"	.200"
	11	12	13	14						
Master/Top Pins	.220"	.240"	.260"	.280"						

* Prior to January 2009 special hollow top pins were used in the control chambers of 6300 series IC cylinders, they were available in sizes 3-11.

Pin Kit 437RC/UL contains these pins.

** UL 437 rated cylinders use stainless steel bottom pins (sizes 1-9 available) in chamber 1 and stainless steel driver/top pins in all chambers.

Notes:

Schlage Classic

76

MACS: 7
Increment: .015"
Progression: Two Step
E.P.D. .500"
Included Angle: 100°
Root Cut: .031"

Key Origination	
Blackhawk	Card 198
CodeMax	DSD #60
HPC Blitz	C45
HPC Punch	P45
ITL	391

Root Depths	
0	.335"
1	.320"
2	.305"
3	.290"
4	.275"
5	.260"
6	.245"
7	.230"
8	.215"
9	.200"

Spacing	
1	.231"
2	.387"
3	.543"
4	.699"
5	.855"
6	1.011"

Notes
1) Read and cut tip-to-bow.
2) Sometimes referred to as "Large Pin" specification.

T.F.C: .231"
B.C.C: .156"

	0	1	2	3	4	5	6	7	8	9
Bottom Pins	.165"	.180"	.195"	.210"	.225"	.240"	.255"	.270"	.285"	.300"
Master Pins			.030"	.045"	.060"	.075"	.090"	.105"	.120"	
Top Pins	.235"				.200"			.165"		

Notes:

Segal Large Pin (Segal Keyway)

77

MACS: 5
Increment: .020"
Progression: Single Step
E.P.D. .485"
Included Angle: 90°
Root Cut: .060"

Key Origination	
Blackhawk	Card 199
CodeMax	DSD #61
HPC Blitz	C46
HPC Punch	
ITL	51

Root Depths	
0	.315"
1	.295"
2	.275"
3	.255"
4	.235"
5	.215"
6	.195"

Spacing	
1	.245"
2	.401"
3	.557"
4	.713"
5	.869"
6	1.025"

T.F.C.: .245"

B.C.C.: .156"

Notes
1) Factory bittings were direct codes written tip-to-bow.
2) Many cylinders were produced with domed pins and beveled plugs producing up to a -.010" root depth variance from official specifications.
3) Segal made many cylinders with a T.F.C. of .262" but retained a B.C.C. of .156". Use HPC Blitz C46, DSD-61, or ITL 395.

	0	1	2	3	4	5	6			
Bottom Pins	.170"	.190"	.210"	.230"	.250"	.270"	.290"			
Master Pins		.020"	.040"	.060"	.080"	.100"	.120"			
Top Pins	.175"									

Notes:

Weiser

78

MACS: 6
Increment: .018"
Progression: Two Step
E.P.D. .483"
Included Angle: 100°
Root Cut: .090"

Key Origination	
Blackhawk	Card 203
CodeMax	DSD #65
HPC Blitz	C50
HPC Punch	
ITL	167

Root Depths	
0	.315"
1	.297"
2	.279"
3	.261"
4	.243"
5	.225"
6	.207"
7	.189"
8	.171"
9	.153"

Spacing	
1	.237"
2	.393"
3	.549"
4	.705"
5	.861"
6	1.017"
7	1.173"

Notes
1) Read and cut bow-to-tip.
2) Used for Falcon multiplex key family.

T.F.C: .237"

B.C.C: .156"

	0	1	2	3	4	5	6	7	8	9
Bottom Pins	.168"	.186"	.204"	.222"	.240"	.258"	.276"	.294"	.312"	.330"
Master Pins			.036"	.054"	.072"	.090"	.108"	.126"	.145"	.160"
Top Pins	.185"	.185"	.185"	.185"	.185"	.185"	.185"	.185"	.185"	.185"

Notes:

Weslock Large Pin

79

MACS: 6
Increment: .0155"
Progression: Two Step
E.P.D. .486"
Included Angle: 100°
Root Cut: .080"

Key Origination	
Blackhawk	
CodeMax	DSD #67
HPC Blitz	C52
HPC Punch	P52
ITL	468

Root Depths	
0	.330"
1	.314"
2	.299"
3	.283"
4	.268"
5	.252"
6	.236"
7	.221"

Spacing	
1	.220"
2	.376"
3	.532"
4	.688"
5	.844"

T.F.C: .220"

B.C.C: .156"

Notes
1) Read and cut bow-to-tip.
2) Original Weslock Large Pin cylinders only use pins 0-7

	0	1	2	3	4	5	6	7	8	9	10
Bottom Pins	.156"	.172"	.1875"	.203"	.219"	.2345"	.250"	.2655"	.281"	.297"	.3125"
Master Pins			.031"	.047"	.0625"	.078"	.094"	.1095"	.125"	.1405"	
Top Pins	.1875"						.219"				

Notes:

Weslock Vanguard

80

MACS: 6
Increment: .0156"
Progression: Two Step
E.P.D. .484"
Included Angle: 100°
Root Cut: .080"

Key Origination	
Blackhawk	
CodeMax	DSD #604
HPC Blitz	
HPC Punch	
ITL	468

Root Depths	
2	.299"
3	.283"
4	.268"
5	.252"
6	.236"
7	.221"
8	.205"
9	.190"
10	.174"

Spacing	
1	.220"
2	.376"
3	.532"
4	.688"
5	.844"
6	.1000"

T.F.C: .220"
B.C.C: .156"

Notes
1) Read and cut bow-to-tip.
2) Factory Vanguard cylinders used pin sizes 2-10.
3) Original Weslock pins were measured in 1/64" increments.

	0	1	2	3	4	5	6	7	8	9	10
Bottom Pins	.1562"	.1718"	.1875"	.2031"	.2187"	.2343"	.250"	.2656"	.2812"	.2968"	.3125"
Master Pins			.0312"	.0468"	.0625"	.0781"	.0937"	.1093"	.125"	.1406"	
Top Pins	.1875"						.2187"				

Notes:

Yale .019"

81

MACS: Varies
Increment: .019"
Progression: Two Step
E.P.D. .502"
Included Angle: Varies
Root Cut: .054"

Key Origination	
Blackhawk	Card 208
CodeMax	DSD #72
HPC Blitz	C57
HPC Punch	
ITL	476

Root Depths	
0	.320"
1	.301"
2	.282"
3	.263"
4	.244"
5	.225"
6	.206"
7	.187"
8	.168"
9	.149"

Spacing	
1	.200"
2	.365"
3	.530"
4	.695"
5	.860"
6	1.025"
7	1.190"

T.F.C: .200"
B.C.C: .165"

Notes
1) Read and cut bow-to-tip.
2) Included angle can be 110°, 95°, or 86° to produce MACS of 4, 5, and 6, respectively.

	0	1	2	3	4	5	6	7	8	9
Bottom Pins	.182"	.201"	.220"	.239"	.258"	.277"	.296"	.315"	.334"	.353"
Master Pins			.038"	.057"	.076"	.095"	.114"	.133"	.152"	
Top Pins	.230"			.190"			.152"			.114"

Notes:

Yale .025"

82

MACS: Varies
Increment: .025"
Progression: Single Step
E.P.D. .502"
Included Angle: Varies
Root Cut: .054"

Key Origination	
Blackhawk	
CodeMax	DSD #79
HPC Blitz	CX9
HPC Punch	
ITL	484

Root Depths	
0	.320"
1	.295"
2	.270"
3	.245"
4	.220"
5	.195"
6	.170"
7	.145"

Spacing	
1	.200"
2	.365"
3	.530"
4	.695"
5	.860"
6	1.025"
7	1.190"

Notes
1) Read and cut bow-to-tip.
2) Included angle can be 110°, 95°, or 86° to produce MACS of 4, 5, and 6, respectively.

T.F.C: .200"
B.C.C: .165"

	0	1	2	3	4	5	6	7
Bottom Pins	.182"	.207"	.232"	.258"	.282"	.307"	.334"	.357"
Master Pins			.050"	.075"	.100"	.125"	.152"	.175"
Top Pins	.230"			.190"			.152"	

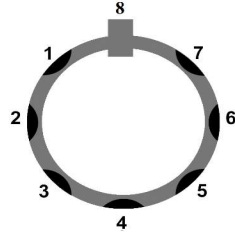
Notes:

Tubular

Chicago ACE/ACE II

84

MACS: N/A
Increment: .0155"
E.P.D. .187"



Cut Depths	
0	-
1	.0155"
2	.0310"
3	.0465"
4	.0620"
5	.0775"
6	.0930"
7	.1085"
8	.1240"

Spacing			
	Standard	Left Offset	Right Offset
1	45°	22.5°	67.5°
2	90°	67.5°	112.5°
3	135°	112.5°	157.5°
4	180°	157.5°	202.5°
5	225°	202.5°	247.5°
6	270°	247.5°	292.5°
7	315°	292.5°	337.5°
8	0°	-	-

Notes
1) Read and cut clockwise from 1 o'clock.
2) Ace and ACE II are available in 7 and 8 pin configurations. 7 pin locks are available in standard, left offset, and right offset configurations.
3) .077" diameter pins. Key pins are paired with driver pins as shown.

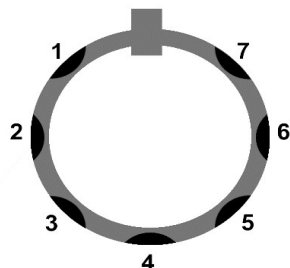
	1	2	3	4	5	6	7			
Key Pins	.2025"	.218"	.234"	.249"	.264"	.279"	.294"			
	1		3		5					
Driver Pins	.180"		.141"		.125"					

Notes:

American Lock (Tubular)

85

MACS: N/A
Increment: .0155"
E.P.D. .188"



Cut Depths	
1	.0155"
2	.0310"
3	.0465"
4	.0620"
5	.0775"
6	.0930"
7	.1085"

Spacing	
1	45°
2	90°
3	135°
4	180°
5	225°
6	270°
7	315°

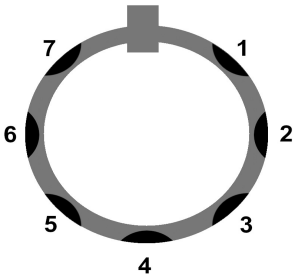
Notes
1) Read and cut clockwise from 1 o'clock.
2) American Lock uses positions 1-7, 8 is blank.
3) Pins are .078" diameter. Each key pin is paired with a driver pin as shown in the chart below.

	1	2	3	4	5	6	7			
Key Pins	.2035"	.2190"	.2345"	.2500"	.2655"	.2810"	.2965"			
	5	5	4	3	2	1	1			
Driver Pins	.200"	.200"	.185"	.170"	.155"	.140"	.140"			

Notes:

Fort GEM (Tubular)

MACS: N/A
 Increment: .0155"
 E.P.D. .185"



Cut Depths	
x	-
0	.0155"
1	.0310"
2	.0465"
3	.0620"
4	.0775"
5	.0930"
6	.1085"
7	.1240"

Spacing	
1	45°
2	90°
3	135°
4	180°
5	225°
6	270°
7	315°

Notes
1) Read clockwise from 1 o'clock. 2) Fort GEM uses positions 1-7, 8 is blank. 3) Pins are .078" diameter. Each key pin is paired with a driver pin as shown in the chart below.

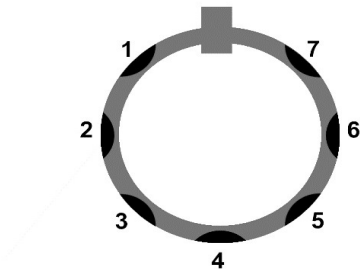
	X	0	1	2	3	4	5	6	7	
Key Pins	.185"	.202"	.218"	.233"	.249"	.264"	.280"	.296"	.311"	
	1									
Driver Pins	.201"									

Notes:

Junkunc Brothers (Old American Lock) (Tubular)

87

MACS: N/A
Increment: .015"
E.P.D. .151"



Cut Depths	
1	.030"
2	.045"
3	.060"
4	.075"

Spacing	
1	45°
2	90°
3	135°
4	180°
5	225°
6	270°
7	315°

Notes
1) Read and cut clockwise from 11 o'clock. 2) American Lock uses positions 1-7, 8 is blank. 3) Pins are .094" diameter.

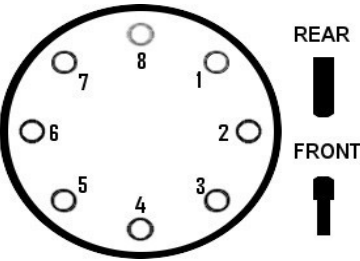
	1	2	3	4						
Key Pins	.180"	.195"	.210"	.225"						
	1									
Driver Pins	.180"									

Notes:

Van Lock 7

88

MACS: N/A
Increment: 0.015"
E.P.D. 0.320"



Key Lengths	
1	.105"
2	.090"
3	.075"
4	.060"
5	.045"
6	.030"
7	.015"
8	.000"
9	.120"

Spacing	
1	45°
2	90°
3	135°
4	180°
5	225°
6	270°
7	315°

Notes
1) Read and assembled clockwise from 1 o'clock.
2) Van Lock 7 uses positions 1-7, 8 is blank.
3) Uses balanced pin stacks with an overall height of 0.435"

	1	2	3	4	5	6	7	8	9	
Front Pins	.215"	.230"	.245"	.260"	.275"	.290"	.305"	.320"	.200"	
Rear Pins	.220"	.205"	.190"	.175"	.160"	.145"	.130"	.115"	.235"	

Notes:

Disc Tumbler/Wafer

Chicago Disc Tumbler

90

MACS: N/A
Increment: .015"
Included Angle: 90°
Root Cut: .035"

Key Origination	
Blackhawk	Card 170
CodeMax	DSD #6
HPC Blitz	C5
HPC Punch	P5L, P5R
ITL	93

Root Depths	
1	.250"
2	.235"
3	.220"
4	.205"
5	.190"

Spacing	
1	.139"
2	.232"
3	.325"
4	.418"
5	.511"
6	.605"

Notes
1) Read and cut bow-to-tip.

T.F.C: .139"
B.C.C: .093"

Notes:

Corbin Disc Tumbler

91

MACS: N/A
Increment: .015"
Included Angle: 90°
Root Cut: .044"

Key Origination	
Blackhawk	Card 173
CodeMax	DSD #12
HPC Blitz	C11
HPC Punch	
ITL	281

Root Depths	
1	.250"
2	.235"
3	.220"
4	.205"
5	.190"

Spacing	
1	.156"
2	.249"
3	.342"
4	.435"
5	.528"
6	.621"

Notes
1) Read and cut bow-to-tip.

T.F.C: .156"
B.C.C: .093"

Notes:

Dexter Disc Tumbler

92

MACS: N/A

Increment: .015"

Included Angle: 90°

Root Cut: .044"

Key Origination	
Blackhawk	Card 152
CodeMax	DSD #5
HPC Blitz	C4
HPC Punch	P5L
ITL	144

Root Depths	
1	.250"
2	.235"
3	.220"
4	.205"
5	.190"

Spacing	
1	.150"
2	.243"
3	.336"
4	.429"
5	.522"

T.F.C: .150"

B.C.C: .093"

Notes
1) Read and cut bow-to-tip.
2) Identical to Briggs & Stratton.

Notes:

Eagle Disc Tumbler

93

MACS: N/A
Increment: .015"
Included Angle: 90°
Root Cut: .044"

Key Origination	
Blackhawk	Card 176
CodeMax	DSD #27
HPC Blitz	See Below
HPC Punch	
ITL	270

Root Depths	
1	.240"
2	.225"
3	.210"
4	.195"
5	.180"

Spacing	
1	.155"
2	.248"
3	.341"
4	.434"
5	.527"

T.F.C: .155"

B.C.C: .093"

Notes
1) Read and cut bow-to-tip.
2) This series used the now discontinued C18 Blitz card.
3) Eagle A63A series (0750 keyway) uses the same depths with a TFC of .090" and BCC of .093".

Notes:

Evinrude-Johnson 73 Series Disc Tumbler

94

MACS: N/A
Increment: .015"
Included Angle: 90°
Root Cut: .044"

Key Origination	
Blackhawk	
CodeMax	
HPC Blitz	
HPC Punch	
ITL	

Root Depths	
1	.247"
2	.232"
3	.217"
4	.202"
5	.187"
6	.172"

Spacing	
1	.130"
2	.225"
3	.320"
4	.415"

T.F.C: .130"

B.C.C: .095"

Notes
1) Read and cut bow-to-tip. 2) Uses Taylor 40F/Ilco OM10.

Notes:

Hudson Disc Tumbler

95

MACS: N/A

Increment: .020"

Included Angle: 90°

Root Cut: .044"

Key Origination	
Blackhawk	Card 181
CodeMax	DSD #40
HPC Blitz	CX32
HPC Punch	PX32
ITL	224

Root Depths	
1	.250"
W or 10	.240"
2	.230"
3	.210"
4	.190"
5	.170"
6	.150"
X or 7	.220"
8	.190"
9	.170"

Spacing	
1	.125"
2	.222"
3	.319"
4	.416"
5	.513"
6	.610"

T.F.C: .125"

B.C.C: .097"

Notes
1) Read and cut bow-to-tip.
2) Applies to EL, GA, GB, GG, HC, HL, HU, LL, MA, MB, MM, W, X series locks. Hudson also makes copies of other manufacturer's locks using original depth and spacing.

Notes:

Hurd Disc Tumbler

96

MACS: N/A

Increment: .020"

Included Angle: 90°

Root Cut: .044"

Key Origination	
Blackhawk	Card 206
CodeMax	DSD# 70
HPC Blitz	C55
HPC Punch	
ITL	238

Root Depths	
1	.250"
2	.230"
3	.210"
4	.190"
5	.170"
7	.220"

Spacing	
1	.125"
2	.220"
3	.315"
4	.410"
5	.505"

T.F.C: .125"

B.C.C: .095"

Notes
1) Read and cut bow-to-tip.

Notes:

Illinois Disc Tumbler (Single Sided)

97

MACS: N/A
Increment: .020"
Included Angle: 90°
Root Cut: .044"

Key Origination	
Blackhawk	Card 187
CodeMax	DSD #44
HPC Blitz	C29
HPC Punch	P29L, P29R
ITL	414

Root Depths	
1	.248"
2	.228"
3	.208"
4	.188"

Spacing	
1	.137"
2	.230"
3	.323"
4	.416"
5	.509"

Notes
1) Read and cut bow-to-tip. 2) Older versions increased the TFC to .150".

T.F.C: Varies

B.C.C: .093"

Notes:

National Disc Tumbler (Single Sided)

98

MACS: N/A
Increment: .025"
Included Angle: 90°
Root Cut: .044"

Key Origination	
Blackhawk	Card 192
CodeMax	DSD #052
HPC Blitz	C37
HPC Punch	P37
ITL	316

Root Depths	
1	.250"
2	.225"
3	.200"
4	.175"

Spacing	
1	.156"
2	.249"
3	.342"
4	.435"
5	.528"
6	.621"

T.F.C: .156"

B.C.C: .093"

Notes
1) Read and cut bow-to-tip.
2) Some factory codes use different depth numbers, substitute as follows: 0 = 1, 2 = 2, 4 = 3, 6 = 4.

Notes:

Schlage Wafer

99

MACS: N/A
Increment: .060"
Progression: N/A
E.P.D. N/A
Included Angle: 90°
Root Cut: .060"

Key Origination	
Blackhawk	
CodeMax	DSD #3360
HPC Blitz	
HPC Punch	
ITL	392

Root Depths	
0	No Cut
1	.235"

Spacing	
1	1.090"
2	.965"
3	.840"
4	.715"
5	.590"
6	.465"
7	.340"
8	.229"

Notes
1) Read and cut tip-to-bow. 2) Index from tip. 3) Combination wafers are matched to each cut on the key. 4) Series wafers are used in each uncut position. 5) Master wafers are only used in the tip position and determine tip cut.

T.F.C.: .275"
B.C.C.: .125"

	1**	2	3	4	5	6	7	8	
Tip*	20	2	4	6	8	0	2	4	Bow*
	10	1	3	5	7	9	1	3	

*Use chart to decode keys/bitting codes. Key is read tip to bow, with raised rib and coining facing you.

**Position 0 is the "series" cut and is determined by the orientation of the plug chambers, this cut extends from the cut center to the end of the key.

**A "W" prefix to the bitting code indicates the use of the W keyway (SC22/1307W blanks), otherwise keys should be cut on a SC6/1307A blank.

**The second digit of a bitting code indicates master keying options. 0 is non-master keyed, 1 is master keyed.

**The remaining 4 digits of the bitting code indicate where cuts will be made, cuts are never directly opposite each other.

**Cut at position 0 is .252" wide extending all the way to the tip of the blank.

**Bitting codes are six digits or six digits preceded by a "W". The first digit is always either 1 or 2 and indicate the series cut.

Notes:

Snap-On Toolbox (CompX Y000)

100

MACS: N/A

Increment: .025"

Included Angle: 90°

Root Cut: .044"

Key Origination	
Blackhawk	
CodeMax	DSD #379
HPC Blitz	
HPC Punch	
ITL	

Root Depths	
1	.290"
2	.265"
3	.240"
4	.215"

Spacing	
1	.130"
2	.225"
3	.320"
4	.415"
5	.510"
6	.605"

T.F.C: .130"

B.C.C: .095"

Notes
1) Read and cut bow-to-tip.
2) Uses CompX D8777 blanks.
3) Existing keys can be duplicated on Ilco 1527 but must be indexed on the center warding grooves.
4) Factory combinations only use depths 2, 3, and 4.

Notes:

Taylor Disc Tumbler

101

MACS: N/A

Increment: .020"

Included Angle: 90°

Root Cut: .044"

Key Origination	
Blackhawk	Card 200
CodeMax	DSD #63
HPC Blitz	C47
HPC Punch	
ITL	411

Root Depths	
1	.210"
2	.190"
3	.170"
4	.150"

Spacing	
1	.098"
2	.193"
3	.288"
4	.383"
5	.478"

T.F.C: .098"

B.C.C: .095"

Notes
1) Read and cut bow-to-tip.
2) Blitz card C47 for this series is now discontinued.

Notes:

Yale Disc Tumbler

102

MACS: N/A

Increment: .020"

Included Angle: 90°

Root Cut: .044"

Key Origination	
Blackhawk	Card 206
CodeMax	DSD# 70
HPC Blitz	C55
HPC Punch	
ITL	238

Root Depths	
1	.250"
2	.230"
3	.210"
4	.190"
5	.170"
7 or X	.220"

Spacing	
1	.125"
2	.220"
3	.315"
4	.410"
5	.505"

T.F.C: .125"

B.C.C: .095"

Notes

1) Read and cut bow-to-tip.

2) Also used by Crosley, Cessna, AMC, Steelcase, BWD, HPC, numerous antique automotive manufacturers, and others.

Notes: